

Functional Critics Are Essential for Actor-Critic: From Off-Policy Stability to Efficient Exploration

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Abstract

The actor-critic (AC) framework has achieved strong empirical success in off-policy reinforcement learning (RL) but suffers from the “moving target” problem, where the evaluated policy changes continually. Functional critics, also known as policy-conditioned value functions, have been proposed to address this issue by explicitly including a representation of the policy as input. While the concept of generalizing value functions across the space of policies is appealing, previous efforts have struggled to remain competitive against standard AC algorithms. In this work, we revisit the concept of functional critics within the actor-critic framework. In particular, we identify two critical aspects that render functional critics a necessity rather than a luxury. First, we demonstrate the power of functional critics in stabilizing the complex interplay between the “deadly triad” and the “moving target”. We provide a convergent off-policy AC algorithm under linear functional approximation that simultaneously dismantles several longstanding barriers between theory and practice: it utilizes target-based TD learning, accommodates dynamic behavior policies, and operates without the restrictive “full coverage” assumption of the behavior policy. By formalizing a dual trust-coverage mechanism, our framework provides principled guidelines for pursuing sample efficiency—rigorously governing the frequency of behavior policy updates and critic re-evaluations to maximize the utility of off-policy data. Second, we uncover a foundational link between functional critics and efficient exploration. We demonstrate that existing model-free approximations of posterior sampling are fundamentally limited in capturing policy-dependent uncertainty, and the functional critic formalism provides the requisite structure to bridge this gap. Each of these results represents, to our knowledge, a first-of-its-kind contribution to the RL literature. Practically, we propose a tailored neural network architecture for the functional critic, along with a minimalist AC algorithm relying solely on these insights. In preliminary experiments on the DeepMind Control Suite, this minimal implementation achieves performance competitive with state-of-the-art methods without requiring standard implementation heuristics.

1 Introduction

Reinforcement learning (RL) has proven to be a powerful tool for sequential decision-making. In this work, we focus on *off-policy* RL under the actor-critic (AC) framework. In contrast to on-policy methods, off-policy methods enjoy improved sample efficiency by reusing past data collected by a behavior policy (Sutton et al., 2011; Lin, 1992; Mnih et al., 2015; Schaul et al., 2016). The actor-critic framework (Degrís et al., 2012; Maei, 2018; Silver et al., 2014; Lillicrap et al., 2015; Wang et al., 2016; Gu et al., 2017; Konda, 2002) is arguably one of the most successful frameworks for policy-based control, which is favored in applications with continuous and/or high-dimensional action spaces, such as robotics and large language models.

Despite the empirical success of off-policy AC, this framework still faces several challenges—both in practice and in theory. Two of the arguably most critical challenges are **training stability** and **efficient exploration**. The latter concerns the issue of acquiring the most useful experience from the environment, while the former concerns how to effectively train the policy given such experience.

On the other hand, the concept of a **Functional Critic** (or Policy-Conditioned Value Function) has been proposed (Harb et al., 2020; Tang et al., 2022; Faccio et al., 2020) to address the difficulty caused by the

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“moving target” problem in AC—where the evaluated policy changes continuously as the actor is updated. The core idea of functional critic is elegant: rather than relearning a transient value function for every new policy, the critic learns a global functional $Q(s, a, \pi)$ that takes the policy representation as an explicit input. Despite the intuitive appeal of functional critics, it remains unclear whether they can effectively address the critical challenges of training stability and efficient exploration. Prior works remain mostly heuristic in arguing the effectiveness of functional critics, and a rigorous understanding of the relation between functional critics and the convergence properties of off-policy AC is missing from the literature. Note that convergence analysis has been explicitly listed under future work in Faccio et al. (2020) and Tang et al. (2022). Furthermore, the relation between functional critics and efficient exploration in off-policy AC is completely underinvestigated in the literature. Consequently, it is not surprising that previous efforts in functional critics have failed to achieve competitive empirical performance against state-of-the-art AC algorithms (Griesbach, 2023).

In this work, we uncover the deeper significance of functional critic modeling. We demonstrate that while functional critics were originally motivated by the “moving target” problem, they are effectively the missing piece required to solve the broader challenges of stability and exploration in off-policy AC.

The Challenge of Training Stability. Stability in off-policy AC is hindered by the complex interplay between the “deadly triad” and the “moving target” problem. Unlike the on-policy setting, applying standard temporal-difference (TD) learning with function approximation to off-policy data can lead to divergence—a phenomenon well-documented as the “deadly triad” (Baird et al., 1995; Tsitsiklis and Van Roy, 1996; Kolter, 2011; Sutton and Barto, 2018). While specialized methods such as gradient TD (Sutton et al., 2009; Qian and Zhang, 2025; Yu, 2017; Maei et al., 2009), emphatic TD (Sutton et al., 2016; Yu, 2015), and target critic-based methods (Mnih et al., 2015; Lee and He, 2019; Zhang et al., 2021; Chen et al., 2023) have been designed to withstand these instabilities, their integration into a provably convergent off-policy AC framework remains a significant challenge. Furthermore, existing guarantees, even for on-policy AC, heavily rely on a multi-timescale design (Konda, 2002), where policy updates must be significantly slower than critic updates.

Historically, three prominent barriers have separated these theoretical results from practical RL recipes.

1. **Algorithmic Complexity:** The only existing convergent off-policy AC (Zhang et al., 2020) relies on a complex hybrid of Gradient TD and Emphatic TD, which introduces significant computational overhead and sensitivities (Manek and Kolter, 2022) in practical settings. In contrast, practitioners favor target-network-based methods (Fujimoto et al., 2018; Haarnoja et al., 2018; Lillicrap et al., 2015) for their stability and ease of tuning, despite a lack of formal convergence proofs under function approximation.
2. **Dynamic Behavior Policies:** Although a convergence result under a changing behavior policy exists for value-based control (Zhang et al., 2021), the corresponding analysis for policy-based control remains largely open¹; existing results (Zhang et al., 2020) typically assume a fixed behavior policy that fails to reflect the evolving rollout policies used in practice.
3. **The Coverage Myth:** Existing off-policy evaluation guarantees generally assume “full coverage” of the behavior policy ($\mu(a|s) > 0$ for all s, a), yet competitive algorithms often employ behavior policies that track the target policy closely, intentionally violating full support to maintain sample efficiency. Conversely, naive choices of behavior policies that easily satisfy full coverage, such as uniform random policies, rarely achieve state-of-the-art results.

Breaking Theoretical Barriers. We overcome these barriers by formalizing the functional critic framework under *linear functional approximation*. This approach serves as a structural generalization of the standard linear value function approximation widely used in literature, extending it to account for the mapping between the policy manifold and the value space (see §3.2).

Our analysis yields the first convergence guarantee for a target-based, off-policy AC algorithm that accommodates both evolving behavior policies and partial coverage. The foundations of this result rest on a novel **off-policy evaluation guarantee under partial coverage** and the development of a **dual trust-coverage mechanism**:

- An **evaluation trust metric** ($\mathcal{C}^{(k)}$) that ensures the behavior policy sufficiently covers the features of the target policy to provide a reliable, high-fidelity anchor for the functional critic.

¹With a notable exception of CAPO (Su et al., 2023) which applies to tabular softmax parameterization.

- A **gradient generalization trust metric** ($\Delta_{k,t}$) that monitors the validity of the anchored critic, rigorously bounding the gradient approximation error as the actor evolves away from the behavior policy.

Together, these criteria provide a rigorous foundation for guaranteed policy improvement in practical, high-dimensional settings where traditional full-coverage assumptions fail. Beyond ensuring stability, this framework establishes principled guidelines for pursuing sample efficiency by governing a nested trade-off: it determines both how long a behavior policy can be utilized for data collection before requiring an update, and how many gradient steps the actor can safely take before the anchored critic must be re-evaluated. By managing these dependencies through a reference policy as the interface between the behavior and target policies, our framework (Algorithm 3) provides a rigorous mechanism to maximize the utility of off-policy data for policy improvement.

Exploration through the Functional Lens. Beyond ensuring training stability, functional critics reveals a foundational link to efficient exploration. Posterior Sampling Reinforcement Learning (PSRL) is a celebrated model-based paradigm inspired by Thompson Sampling (for multi-arm bandits), yet it remains computationally prohibitive in complex, high-dimensional environments. While existing model-free approximations of PSRL (Osband et al., 2016, 2018, 2019) attempt to capture uncertainty via ensembles or randomized priors, we demonstrate their fundamental limitations in accounting for policy-dependent uncertainty. We illustrate that the functional critic formalism is the requisite tool to bridge this gap. By explicitly representing the mapping between the policy manifold and the value space, our framework enables a true posterior sampling style of exploration within the actor-critic architecture—positioning the functional critic as a unified solution for both training stability and principled exploration.

From Theory to Practice. To validate these insights, we develop a concrete implementation of the functional critic that leverages the expressive power of modern neural architectures (Vaswani et al., 2017; Radford et al., 2018, 2019; Dosovitskiy et al., 2020; Devlin et al., 2019). In §4, we present a minimalist implementation designed to isolate the effects of our core stability and exploration principles. Notably, we purposefully omit the suite of common heuristics—such as minimum twin-target values, added exploration noise, or (automated) entropy regularization—that are typically considered essential for competitive performance in deep RL. Despite this minimalism, our method outperforms competitive state-of-the-art off-policy AC baselines on representative continuous control tasks from the DeepMind Control Suite (Tassa et al., 2018). While these results are currently limited to specific benchmark domains, the consistent performance gains achieved without standard heuristic tuning underscore the inherent robustness of the functional critic framework and its potential to serve as a new, theoretically grounded standard for stable and efficient reinforcement learning.

Summary of contributions. This work makes three primary contributions:

- 1. Stability and Convergence via Functional Critics:** We dismantle the longstanding barriers between off-policy AC theory and practice by establishing the first convergence proof for target-based AC under linear functional approximation. Moving beyond the restrictive “full coverage” assumptions and fixed behavior policies, we establish a dual trust-coverage mechanism that provides principled guidelines for pursuing sample efficiency. This framework rigorously governs the frequency of both behavior policy updates and critic re-evaluations, offering a formal mechanism to maximize the utility of off-policy data for policy improvement.
- 2. Exploration through the Functional Lens:** We reveal a foundational link between functional critics and efficient exploration. By characterizing the fundamental limitations of existing model-free approximations of Posterior Sampling Reinforcement Learning (PSRL), we illustrate that the functional critic formalism—specifically its ability to capture the mapping between the policy manifold and the value space—is a structural requirement for rigorous model-free posterior sampling. This positions our framework as a unified tool that not only ensures stability but serves as the essential architecture for principled, uncertainty-aware exploration in high-dimensional settings.
- 3. Principles-First Implementation:** Translating our theoretical framework into a practical deep RL recipe, we propose a tailored neural architecture and a minimalist algorithm. In representative continuous

control tasks from the DeepMind Control Suite, our implementation achieves consistent performance gains over state-of-the-art baselines, despite intentionally omitting standard implementation heuristics.

2 Preliminaries

Markov Decision Process. We consider an infinite-horizon *Markov Decision Process* (MDP) defined by a finite state space $\mathcal{S}$ with $|\mathcal{S}|$ states, a finite action space $\mathcal{A}$ with $|\mathcal{A}|$ actions², a transition kernel $p : \mathcal{S} \times \mathcal{S} \times \mathcal{A} \rightarrow [0, 1]$, a reward function $r : \mathcal{S} \times \mathcal{A} \rightarrow [0, 1]$, and a discount factor $\gamma \in [0, 1]$. At each time-step t , after an action a_t is picked, agent then proceeds to a new state s_{t+1} according to $p(\cdot|s_t, a_t)$ and gets a reward $r(s_t, a_t)$. Given any policy $\pi : \mathcal{S} \times \mathcal{A} \rightarrow \Delta(\mathcal{A})$, we define

$$V_\pi(s) := \mathbb{E}\left[\sum_{t=0}^{\infty} \gamma^t r(s_t, \pi(s_t)) \mid s_0 = s\right],$$

$$Q_\pi(s, a) := r(s, a) + \gamma \mathbb{E}_{s_1} [V_\pi(s_1) \mid s_0 = s, a_0 = a],$$

where the expectations are taken with respect to all future transitions and the randomness of π .

Off-Policy Reinforcement Learning. In off-policy RL, the learner lacks knowledge of the underlying transition kernel p , but can interact with the environment through a *behavior policy* μ . The objective is to identify the optimal policy π^* that maximizes the expected return $J_\rho(\pi) := \sum_s \rho(s) V_\pi(s)$ for some fixed distribution ρ over initial states.³ We parameterize the policy space by $\theta \in \Theta \subseteq \mathbb{R}^d$ and denote the value and action-value functions under policy π_θ as V_θ and Q_θ , respectively.

The Actor-Critic Framework. The actor-critic framework consists of two key components: the *actor*, which selects actions according to a parameterized policy π_θ , and the *critic*, which evaluates this policy by estimating V_θ or Q_θ . For training, policy gradient methods are commonly employed to update the actor where the critic is first updated and then used to evaluate policy gradients, as detailed below:

(a) Critic update. One core challenge in the critic’s update step arises from the instability of temporal-difference (TD) learning when combined with function approximation in the off-policy setting—a phenomenon known as the *deadly triad*. More precisely, to evaluate a given policy π , while TD learning is effective in on-policy settings, it is provably unstable in off-policy setting when function approximation are used due to the mismatch of d_μ and d_π . This instability has motivated a line of research on convergent methods for off-policy evaluation, including gradient-based TD (Sutton et al., 2009; Zhang and Whiteson, 2022; Qian and Zhang, 2025), Emphatic TD (Sutton et al., 2016; Yu, 2015), and target-network based methods (Chen et al., 2023; Zhang et al., 2021; Fujimoto et al., 2018; Lillicrap et al., 2015).

(b) Actor update. In the on-policy setting, actor update can leverage the policy-gradient theorem (Sutton et al., 2000),

$$\nabla_\theta J_\rho(\theta) \propto \sum_{s \in \mathcal{S}} d_\pi^\rho \sum_{a \in \mathcal{A}} \nabla_\theta \pi_\theta(a|s) Q_\theta(s, a),$$

where d_π^ρ is the stationary state distribution under π with starting state following ρ . For off-policy settings, however, it is not possible to sample from d_π^ρ . Instead, directly following the chain rule gives the following formula,

$$\nabla_\theta J_\rho(\theta) = \sum_{s \in \mathcal{S}} \rho(s) \sum_{a \in \mathcal{A}} \left[\nabla_\theta \pi_\theta(a|s) Q_\theta(s, a) + \pi_\theta(a|s) \nabla_\theta Q_\theta(s, a) \right]. \quad (2.1)$$

The difficulty lies in the second term of the summand of (2.1), $\pi_\theta(a|s) \nabla_\theta Q_\theta(s, a)$, which is in general hard to compute. With the functional critic, similar as in Faccio et al. (2020), (2.1) can be directly estimated,

$$\nabla_\theta J_\rho(\pi_\theta; \xi) = \mathbb{E}_{s \sim \rho} \left[\sum_{a \in \mathcal{A}} \hat{Q}(\pi_\theta, s, a; \xi) \nabla_\theta \pi_\theta(a|s) + \pi_\theta(a|s) \nabla_\theta \hat{Q}(\pi_\theta, s, a; \xi) \right], \quad (2.2)$$

²We assume finite states and actions just for simplicity of theoretical exposition. However, our algorithm design remains applicable to infinite spaces, and our theoretical results can be extended to settings with countable s and a .

³Existing off-policy AC results typically assume a fixed behavior policy μ and a surrogate objective with $\rho = d_\mu$, where d_μ denotes the stationary state distribution under μ .

where $\hat{Q}(\pi_\theta, s, a; \xi)$ is the functional critic parameterized by ξ .

An alternative exact off-policy gradient formula for the surrogate objective J_{d_μ} was proposed by Imani et al. (2018),

$$\nabla_\theta J_{d_\mu}(\theta) = \sum_s m_\theta(s) \sum_a \nabla_\theta \pi_\theta(a|s) Q_\theta(s, a),$$

where $m_\theta(s) := \mathbb{E}_{a_t \sim \pi_\theta(s_t)} [\sum_{t=0}^{+\infty} d_\mu(s_t) | s_0 = s]$ is a policy-dependent *emphasis* term, which reflects state dependent reweighting to correct the off-policyness. Just like the value function, this term needs to be estimated from data, and is similarly prone to the instability issues mentioned in §1. This additional estimation step requires extra learning-rate schedules, which further complicates hyperparameter tuning and prevents the method from scaling as effectively as simpler approaches.

Posterior Sampling for Reinforcement Learning (PSRL). PSRL is inspired by a classical Bayesian approach to the multi-arm bandit problem known as Thompson Sampling (TS). TS maintains a posterior of the expected reward for each arm and selects an arm according to the probability that it is the optimal one. This algorithm achieves optimal Bayesian regret by ensuring efficient exploration while minimizing long-term regret. PSRL extends this principle to MDPs: maintaining a posterior over the transition and reward models. It operates in an episodic fashion: at the start of each episode, it samples a single MDP from the posterior, computes the optimal policy for the sampled MDP, and follows this policy for the whole episode. The collected experience is then used to update the posterior over MDP. While being appealing both theoretically and empirically in tabular settings (Osband et al., 2013), PSRL’s reliance on model-based sampling often becomes computationally prohibitive in high-dimensional continuous spaces.

3 Methodology

In this section, we develop our off-policy actor-critic framework and establish its theoretical foundations. We begin in §3.1 by introducing a general meta-algorithm that leverages function approximation for both the actor and the functional critic. This high-level framework serves as the blueprint for our subsequent analysis and implementation.

In §3.2, we present our primary theoretical results under linear functional approximation, dismantling several longstanding barriers in the analysis of off-policy training stability. We first establish a target-based policy evaluation guarantee under a relaxed, feature-based coverage condition. The key technical innovation lies in characterizing the **Bellman contraction under a carefully designed L^2 -norm weighted by the on-policy feature covariance**.

This analytical tool provides a rigorous foundation for our first trust-coverage criterion, which quantifies the reliability of policy evaluation without requiring full feature coverage from behavior policies. Building on this, we derive a second criterion for gradient estimation reliability by leveraging both the evaluation guarantee and the generalization properties of the functional critic representation. Together, these criteria form a **dual-layer trust-coverage** mechanism for guaranteed policy improvement.

Beyond merely accommodating dynamic behavior policies, this hierarchical framework (Algorithm 3) establishes **principled guidelines for pursuing sample efficiency** by governing a nested trade-off. Specifically, the first criterion determines how long a behavior policy can be utilized for data collection before requiring an update to track the reference policy, while the second criterion dictates how many gradient steps the actor can safely take before the anchored critic must be re-evaluated with an updated reference policy. By managing these dependencies through the reference policy as the interface between the behavior and target policies, our approach provides a rigorous mechanism to maximize the utility of off-policy data.

Finally, §3.3 examines the requirements for efficient exploration. We characterize the fundamental limitations of existing model-free approximations of Posterior Sampling Reinforcement Learning (PSRL) and illustrate how the functional critic formalism serves as the requisite tool to enable true posterior sampling-style exploration in model-free settings.

3.1 Meta Algorithm: Functional Actor Critic

The overall procedure of the functional critic framework is presented as a meta-algorithm composed of modular sub-routines for policy evaluation (corresponding to the functional critic update) and policy improvement

Algorithm 1 Functional Actor-Critic Meta Algorithm

- 1: **Inputs:** Initial actor parameter θ_0 , initial functional critic parameter ξ_0 , number of epochs T , batch size m , actor update step-size schedule $\{\eta_t\}_{t=1}^T$
 - 2: **for** $t = 1, \dots, T$ **do**
 - 3: Sample $\mathcal{D}_t \leftarrow (s_t, a_t, r_t, s'_t)$ from behavior policy.
 - 4: Update functional critic parameter $\xi_t \leftarrow$ **Functional Policy Evaluator** $(t, \xi_{t-1}, \theta_{t-1}, \mathcal{D}_t)$
 - 5: Compute the **Parameterized Off-Policy Gradient** $G_t \leftarrow \nabla_{\theta}(\mathcal{J}(\pi_{\theta}; \xi_t))|_{\theta=\theta_{t-1}}$ // See (2.2)
 - 6: Update the actor parameter $\theta_t \leftarrow \theta_{t-1} - \eta_t G_t$.
 - 7: **end for**
 - 8: **Return** θ_T
-

(corresponding to the actor update), as summarized in Algorithm 1.

Functional Policy Evaluator. The critic is defined as a trainable functional $\hat{Q}(\cdot; \xi)$, parameterized by ξ , which takes as input a triplet $(\pi_{\theta}, s, a) \in \Theta \times \mathcal{S} \times \mathcal{A}$ and aims to approximate the policy value $Q_{\theta}(s, a)$. This further gives the functional value evaluator

$$\mathcal{J}(\pi_{\theta}; \xi) := \sum_{a,s} \pi_{\theta}(a|s) d_{\mu}(s) \hat{Q}(\pi_{\theta}, s, a; \xi).$$

The motivation behind functional evaluator is to decouple policy optimization from value estimation while enabling the critic to generalize across time-varying input policies.

Following this insight, the policy evaluation of changing policies can be formulated as a continual TD-learning problem over *functionals*, where the learning objective at time step t is to match the Bellman equation for the policy π_{θ_t} ,

$$\hat{Q}(\pi_{\theta_t}, s, a; \xi) \approx r_{\pi_{\theta_t}}(s, a) + \gamma \mathbb{E}_{s'} [\sum_{a'} \pi_{\theta_t}(a'|s') \hat{Q}(\pi_{\theta_t}, s', a'; \xi) | s, a], \quad (3.1)$$

whereas the expectation over s' can be estimated from data.

Parametrized Off-Policy Gradient. The actor update sub-routine in our framework follows the off-policy policy gradient formula (2.2), where the exact off-policy gradient can be derived directly from the learned functional critic $\hat{Q}(\pi_{\theta_t}, s, a; \xi)$.

3.2 Convergence Analysis under Linear Functional Approximation

In this section, we characterize the theoretical advantages of the functional critic framework by analyzing a specific instantiation of Algorithm 1 under **linear functional approximation**. We focus on a target-based AC design to demonstrate how the functional perspective enables convergence guarantees that were previously unattainable under practical off-policy conditions—specifically in settings with evolving behavior policies and partial feature coverage. To maintain clarity in this exposition, we present the key technical results and their implications here, while providing complete proofs for all theorems in Appendix B.

3.2.1 Linear Functional Approximation Setup

We assume the following regarding the MDP dynamics and the structure of the value function space.

Assumption 1 (Strong Mixing Condition). *For any policy $\pi \in \Pi$, the induced Markov chain $\{s_k\}$ is irreducible and aperiodic, ensuring a unique stationary distribution d^{π} .*

Assumption 2 (Linear Functional Representation). *For the policy class Π , there exists a feature map $\phi : \Pi \times \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{B}^d$ and a vector $\xi^* \in \mathbb{R}^d$ with $\|\xi^*\|_2 \leq L$ such that $Q^{\pi}(s, a) = \langle \phi^{\pi}(s, a), \xi^* \rangle$, for all $(\pi, s, a) \in \Pi \times \mathcal{S} \times \mathcal{A}$.*

Assumption 3 (Bellman Completeness). *The linear functional class is approximately closed under the Bellman operator $\mathcal{T}^\pi$. Specifically, for any $\xi \in \mathbb{B}^d$, there exists some $\xi' \in \mathbb{B}^d$ such that for all $(\pi, s, a) \in \Pi \times \mathcal{S} \times \mathcal{A}$:*

$$\langle \phi^\pi(s, a), \xi' \rangle = \mathcal{T}^\pi([\langle \phi^\pi(\cdot, \cdot), \xi \rangle])(s, a) + \mathcal{E}_{approx},$$

where $[\cdot]$ denotes the projection onto the value range $[0, \frac{1}{1-\gamma}]$, and $\mathcal{E}_{approx}$ is the inherent approximation error.

Assumption 1 is a standard ergodicity condition required for stochastic approximation with Markovian noise, consistent with established off-policy RL literature (Zhang et al., 2020, 2021; Chen et al., 2023). While some works (Zhang et al., 2020, 2021) omit aperiodicity for a slightly weaker condition, our inclusion of it ensures fast mixing and is standard for providing a more direct path to practical convergence rates.

Assumption 2 provides a realizability condition for the functional class. This generalizes standard linear representations used in literature (Zhang et al., 2020, 2021; Chen et al., 2023; Duan et al., 2020; Xiong et al., 2022) to functional features that explicitly encode the mapping between the policy manifold Π and the value space. Unlike static representations that assume a fixed basis for a single policy, our functional approach treats the value as a smooth map across the intrinsic geometry of Π . We provide a detailed geometric perspective in Appendix B.1, illustrating how Assumption 2 arises naturally from a first-order Taylor expansion on the Riemannian policy manifold, where the functional features represent displacements on the local tangent space.

Similarly, Assumption 3 generalizes approximate Bellman completeness (e.g. Duan et al. (2020)) to the functional setting. We note that this completeness assumption is weaker and more flexible than the "Linear MDP" requirements made in many previous works (Xiong et al., 2022; Yin et al., 2022; Min et al., 2021), as it does not demand the transition dynamics themselves to be linear in the features.

3.2.2 Target-Based Evaluation with Partial Coverage

We first analyze the policy evaluation subroutine detailed in Algorithm 2, which estimates the value of a target policy π using samples collected from a behavior policy μ . This design introduces an auxiliary "target network" $\hat{\xi}$ to anchor the Q -function prediction, following a standard temporal-difference subroutine that incorporates truncation for stability (Chen et al., 2023).

Algorithm 2 Target-Network Based Policy Evaluation

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1: Input: Behavior policy  $\mu$ ; target policy  $\pi$ ; feature map  $\phi^\pi$ ;
   inner-loop error level  $\epsilon$ ; inner iterations  $K$ ; outer iterations  $T$ .
2: Initialize: weights  $\{\xi_{t,0}\}_{t=1}^T \leftarrow \mathbf{0}$ ,  $\hat{\xi}_0 \leftarrow \mathbf{0}$ .
3: for  $t = 0, 1, \dots, T-1$  do
4:   for  $k = 0, 1, \dots, K-1$  do
5:     Sample transition  $(s_k, a_k, r_k, s'_k)$  from  $\mu$ 
6:     Compute TD target:  $y_k \leftarrow r_k + \gamma \sum_a \pi(a|s'_k) \cdot [\phi^\pi(s'_k, a)^\top \hat{\xi}_t]$ 
7:     Update weights:  $\xi_{t,k+1} \leftarrow \xi_{t,k} + \alpha \phi^\pi(s_k, a_k)(y_k - \phi^\pi(s_k, a_k)^\top \xi_{t,k})$ 
8:   end for
9:    $\hat{\xi}_{t+1} \leftarrow \xi_{t,K}$  {Update target network}
10: end for
11: Output: Final estimate  $\hat{\xi}_T$ .
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While convergence for such procedures under linear function approximation is well documented (Carvalho et al., 2020; Chen et al., 2023; Zhang et al., 2021), existing results typically rely on a "full coverage" assumption of the behavior policy: $\mu(a|s) > 0$ whenever $\pi(a|s) > 0$. This condition can be restrictive in high-dimensional or continuous action spaces. In fact, this restriction on μ persists across almost all TD-type algorithms.

To dismantle this barrier, we demonstrate that the target-based design enables convergence under a weaker notion of **feature-based coverage**. Specifically, stability is guaranteed as long as the behavior policy provides sufficient coverage over the directions in feature space relevant to the target policy π , even in the absence of point-wise action support. We formalize this requirement through the following metric, which quantifies the alignment between the target features and the empirical covariance of the behavior distribution.

Definition 1 (Evaluation Trust Metric). For a target policy π and a behavior policy μ , let $\tilde{\mathbf{A}}_\pi := \mathbb{E}_{d^\mu}[\phi^\pi(s, a)\phi^\pi(s, a)^\top]$ be the feature covariance matrix of π under the behavior distribution. The evaluation trust metric $\mathcal{C}(\pi, \mu)$ is defined as:

$$\mathcal{C}(\pi, \mu) := \mathbb{E}_{\bar{d}_\pi} [\|\phi^{(k)}(s, a)\|_{\tilde{\mathbf{A}}_\pi^{-1}}^2], \quad (3.2)$$

where $\|\mathbf{u}\|_{\mathbf{A}^{-1}} := \lim_{\lambda \rightarrow 0^+} \|\mathbf{u}\|_{(\mathbf{A} + \lambda \mathbf{I})^{-1}}$ for a positive semi-definite matrix $\mathbf{A}$. $\bar{d}_\pi$ denotes the stationary measure over $\mathcal{S} \times \mathcal{A}$ under π .

Following the definition of our coverage metric, we now establish the formal convergence guarantee. As highlighted previously, the key to dismantling the full-coverage barrier lies in characterizing the Bellman contraction under an L^2 -norm weighted by the on-policy feature covariance, $\mathbf{A}_\pi := \mathbb{E}_{d^\pi}[\phi^\pi(s, a)\phi^\pi(s, a)^\top]$. By measuring errors specifically within this manifold, we ensure that the behavior policy only needs to provide coverage in the directions relevant to the target policy’s evolution.

Theorem 3.1 (Evaluation Guarantee with Partial Coverage). Let Assumptions 1–3 hold. If the parameters K, α in Algorithm 2 are suitably selected (with $K \asymp \epsilon^{-2}$), then the estimated functional critic $\hat{Q}_T$ satisfies:

$$\mathbb{E}[\|\hat{Q}_T(s, a) - Q^\pi(s, a)\|_{d^\pi}^2] \lesssim \gamma^T + \mathcal{C}(\pi, \mu)(\epsilon^2 + \mathcal{E}_{\text{approx}}^2) + \mathcal{E}_{\text{approx}}^2.$$

where $\|\cdot\|_{d^\pi}^2$ denotes the L^2 -norm weighted by the on-policy distribution.

Remark 1 (Results under Standard Linear Function Approximation). Theorem 3.1 requires Assumptions 2 and 3 to hold only for the specific policy π under evaluation, and thus it automatically applies under the standard linear function approximation setting (Carvalho et al., 2020; Chen et al., 2023; Zhang et al., 2021). As long as $\mathcal{C}(\pi, \mu) < \infty$, this bound matches the statistical efficiency of least-squares evaluation approaches (Min et al., 2021; Duan et al., 2020).

Theorem 3.1 represents, to our knowledge, the first evaluation guarantee for a TD-type algorithm that replaces point-wise full coverage conditions with a flexible feature-based requirement. This is made possible by the on-policy weighting, which allows the analysis to remain valid even when the off-policy covariance $\tilde{\mathbf{A}}_\pi$ is singular (utilizing the Moore-Penrose pseudoinverse convention), provided the features required by π are contained within the span of the data collected by μ .

This stability in value space translates directly to the parameter space, which we utilize to analyze our full actor-critic algorithm in the following section.

Corollary 1 (Convergence in Weight Space). The convergence of the weights $\hat{\xi}_T$ toward the true parameters ξ^* is bounded under the on-policy feature covariance $\mathbf{A}_\pi$:

$$\mathbb{E}[\|\hat{\xi}_T - \xi^*\|_{\mathbf{A}_\pi}^2] \lesssim \gamma^T + \mathcal{C}(\pi, \mu)(\epsilon^2 + \mathcal{E}_{\text{approx}}^2) + \mathcal{E}_{\text{approx}}^2.$$

3.2.3 Off-Policy AC with the Dual-Layer Trust-Coverage Mechanism

This section marks the transition from static policy evaluation to the full off-policy linear functional actor-critic (Algorithm 3). By leveraging the stable parameter estimates from the previous section, we establish the **dual-layer trust-coverage** criteria as a hierarchical control mechanism for policy improvement.

Notation Shorthand. To maintain clarity throughout the analysis, we denote the policy $\pi_{\theta_t^{(k)}}$ simply as $\pi_t^{(k)}$ for the remainder of this section and the next. Correspondingly, the functional features evaluated at the current policy are denoted by $\phi_t^{(k)}(s, a) := \phi(s, a; \theta_t^{(k)})$. We further simplify the notation for covariance matrices by denoting the reference (on-policy) feature covariances $\mathbf{A}_{\pi^{(k)}}$ as $\mathbf{A}_k$. Note that $\mathbf{A}_k$ remains fixed throughout epoch k , serving as the metric anchor for both trust-coverage metrics.

Algorithm 3 employs an epoch-based refinement schedule that prioritizes sample efficiency by maximizing the utility of behavior policy data. At the start of each epoch k , the behavior policy μ is dynamically determined by the **evaluation trust metric** $\mathcal{C}(\pi^{(k)}, \mu)$: if the current data distribution fails to provide sufficient coverage for current reference policy ($\mathcal{C}(\pi^{(k)}, \mu) > C_{\text{max}}$), the behavior policy is updated to match the reference ($\mu \leftarrow \pi^{(k)}$) to restore evaluation stability. Once coverage is guaranteed, the target-network-based evaluation (Algorithm 2) is invoked to obtain the anchored estimator $\xi^{(k)}$, which provides the gradient signal

Algorithm 3 Feature-Based Functional Actor–Critic

```
1: Input: feature map  $\phi$ ; accuracy level  $\epsilon$ ; stepsize  $\eta$ ; evaluation trust metric bound  $C_{\max} \geq 1$ 
2: Initialize: reference policy  $\bar{\pi}^{(0)}$ , behavior policy  $\mu \leftarrow \bar{\pi}^{(0)}$ ,
3: {EPOCH-WISE REFINEMENT}
4: for  $k = 0, 1, \dots$  do
5:   Compute evaluation trust metric  $\mathcal{C}(\bar{\pi}^{(k)}, \mu)$  via (3.2)
6:   if  $\mathcal{C}(\bar{\pi}^{(k)}, \mu) > C_{\max}$  then
7:     Update behaviour policy to reference  $\mu \leftarrow \bar{\pi}^{(k)}$  {Restore coverage stability}
8:   end if
9:   Policy Evaluation: compute  $\xi^{(k)}$  via Algorithm 2 using  $(\mu, \bar{\pi}^{(k)}, \phi, \epsilon = \epsilon)$ 
10:  Reset target policy:  $\pi_0^{(k)} \leftarrow \bar{\pi}^{(k)}$ 
11:  {POLICY IMPROVEMENT LOOP}
12:  for  $t = 1, 2, \dots$  do
13:    Improvement step:  $\pi_t^{(k)} \leftarrow \pi_{t-1}^{(k)} + \eta \hat{\nabla} J(\pi_{t-1}^{(k)}, \xi^{(k)})$ 
14:    Compute gradient generalization trust metric  $\Delta_{k,t}$  via (3.4)
15:    if  $\Delta_{k,t} > 2$  then
16:      Update reference policy for next epoch:  $\bar{\pi}^{(k+1)} \leftarrow \pi_t^{(k)}$ 
17:      break {Terminate epoch to refresh anchor}
18:    end if
19:  end for
20: end for
21: Output: Final policy  $\pi$ 
```

for the subsequent policy improvement loop. As the actor $\pi_t^{(k)}$ evolves, the **gradient generalization trust metric** $\Delta_{k,t}$ (formally defined in (3.4)) monitors the validity of the anchored critic. If $\Delta_{k,t}$ violates the threshold, the reference policy is updated to match the current actor and the epoch terminates, triggering a re-assessment of the evaluation trust metric and a refresh of the functional critic for the next epoch.

Remark 2 (Similarity-Based Generalization). *The ability to compute gradients based on policy similarity is a direct consequence of the functional representation perspective. In the limiting case where the effective epoch length equals one—implying that successive policies share no exploitable similarity—our method naturally recovers a standard two-timescale off-policy AC scheme. Thus, our framework gracefully recovers classical results while offering a more efficient, multi-step improvement path through functional generalization.*

Remark 3 (Selection of Behavior Policies). *Standard selections of the behavior policy in Algorithm 3 recover the following frameworks as special cases:*

- **On-Policy Adaptation:** Fixing $\mu = \bar{\pi}^{(k)}$ throughout each epoch satisfies the coverage conditions by construction, recovering the on-policy AC framework of Barakat et al. (2022) while still benefiting from functional generalization.
- **Off-Policy Stability:** Under traditional full coverage ($\mu(a|s) > 0$), our framework functions as a stable off-policy AC algorithm but with the added efficiency gains of functional critic modeling.

3.2.4 Convergence Analysis of Algorithm 3

Since the convergence of perturbed policy gradient methods under ϵ -accurate gradient estimates is well-established (Jain et al., 2017; Zhang et al., 2020; Barakat et al., 2022), our analysis focuses on demonstrating that Algorithm 3 maintains the gradient estimation error below a prescribed threshold ϵ . To ensure this accuracy, we must control two distinct error sources:

Reference Evaluation Error: The accuracy of the functional $\xi^{(k)}$ in evaluating the reference policy $\bar{\pi}^{(k)}$.

Gradient Generalization Error: The gradient error incurred as the actor $\pi_t^{(k)}$ drifts from the reference $\bar{\pi}^{(k)}$.

Layer 1: Reference Policy Evaluation

Our evaluation results in Theorem 3.1 provide a sufficient conditions for the stability of the functional anchor. Specifically, as long as the behavior policy μ satisfies the coverage condition $\mathcal{C}(\bar{\pi}^{(k)}, \mu) \leq C_{\max}$, the estimated functional critic achieves the following error guarantee (cf. Corollary 1):

$$\sqrt{\mathbb{E}[\|\xi^{(k)} - \xi^*\|_{\mathbf{A}_k}^2]} \lesssim \bar{\epsilon} := \sqrt{C_{\max}} \cdot (\epsilon + \mathcal{E}_{\text{approx}}). \quad (3.3)$$

Here, the on-policy feature covariance $\mathbf{A}_k$ is anchored at the reference policy $\bar{\pi}^{(k)}$.

Layer 2: Gradient Generalization and the Second Trust Metric.

Under linear functional approximation, the off-policy gradient error $\nabla J(\pi_t^{(k)}) - \hat{\nabla} J(\pi_t^{(k)}, \xi^{(k)})$ decomposes into two primary components:

$$\underbrace{\mathbb{E}_{s \sim \rho} \left[\sum_{a \in \mathcal{A}} \langle \phi_t^{(k)}(s, a), \xi^{(k)} - \xi^* \rangle \nabla \pi_t^{(k)}(a|s) \right]}_{:= \mathcal{G}_1^{(k,t)}} + \underbrace{\mathbb{E}_{s \sim \rho} \left[\sum_{a \in \mathcal{A}} \pi_t^{(k)}(a|s) D \phi_t^{(k)}(s, a) (\xi^{(k)} - \xi^*) \right]}_{:= \mathcal{G}_2^{(k,t)}}$$

To bound $\mathcal{G}_1$ and $\mathcal{G}_2$ using (3.3), we introduce the **Gradient Generalization Trust Metric** $\Delta_{k,t}$, which quantifies the deviation of current policy $\pi_t^{(k)}$ from the reference along the directions of the features ϕ and their gradients $\nabla \phi$:

$$\Delta_{k,t} := \max\{\mathcal{C}_1^{(k,t)}, \mathcal{C}_2^{(k,t)}\}, \quad (3.4)$$

where the component coefficients are defined as:

$$\begin{aligned} \mathcal{C}_1^{(k,t)} &= \left\| \mathbb{E}_{s \sim \rho} \left[\sum_{a \in \mathcal{A}} \nabla \pi_t^{(k)}(a|s) \|\phi_t^{(k)}(s, a)\|_{\mathbf{A}_k^{-1}} \right] \right\|_{\infty}, \\ \mathcal{C}_2^{(k,t)} &= \mathbb{E}_{s \sim \rho} \left[\sum_{a \in \mathcal{A}} \pi_t^{(k)}(a|s) \|D \phi_t^{(k)}(s, a)\|_{\mathbf{A}_k^{-1}, \infty} \right]. \end{aligned}$$

By Cauchy-Schwarz inequality, $\|\mathcal{G}_1^{(k,t)}\|_{\infty} \leq \mathcal{C}_1^{(k,t)} \|\xi^{(k)} - \xi^*\|_{\mathbf{A}_k}$, and $\|\mathcal{G}_2^{(k,t)}\|_{\infty} \leq \mathcal{C}_2^{(k,t)} \|\xi^{(k)} - \xi^*\|_{\mathbf{A}_k}$. Applying (3.3) and (3.4), then the expected gradient error is controlled by $\Delta_{k,t} \bar{\epsilon}$. The $\Delta_{k,t}$ -based criterion in Algorithm 3 thus provides a principled mechanism for terminating the epoch and refreshing the functional anchor.

By monitoring the dual-layer trust metrics, we ensure the gradient signal remains dominant throughout training. The following theorem establishes the convergence of Algorithm 3 under the standard regularity conditions, which simply states the continuity of ϕ on θ (Zhang et al., 2020, 2021; Barakat et al., 2022):

Assumption 4 (Smoothness). *There exists some $L > 0$ so that for any $\theta, \theta' \in \Theta$ and corresponding $\pi_{\theta}, \pi_{\theta'}$,*

$$\|\phi^{\pi_{\theta}}(s, a) - \phi^{\pi_{\theta'}}(s, a)\|_2 \leq L \|\theta - \theta'\|_2, \quad \|D \phi^{\pi_{\theta}}(s, a) - D \phi^{\pi_{\theta'}}(s, a)\|_2 \leq L \|\theta - \theta'\|_2, \quad \forall (s, a) \in \mathcal{S} \times \mathcal{A}.$$

Theorem 3.2 (Convergence of Functional Actor-Critic). *Let Assumptions 1–3 hold. For a suitable stepsize η , the sequence of policies $\{\pi_m\}$ generated by Algorithm 3 over M total steps, including K episodes and T_k time-step at k -th episode such that $\sum_{k=1}^K T_k = M$, satisfies*

$$\frac{1}{M} \mathbb{E} \left[\sum_{k=1}^K \sum_{t=1}^{T_k} \|\nabla J_{\rho}(\pi_t^{(k)})\|^2 \right] \leq \frac{2(J_{\rho}^* - J_{\rho}(\pi^{(0)}))}{\eta M} + \mathcal{O}(\bar{\epsilon}^2), \quad (3.5)$$

where $\bar{\epsilon} := \sqrt{C_{\max}} \cdot (\epsilon + \mathcal{E}_{\text{approx}})$ represents the reference evaluation error from (3.3).

Interpretation: Stability via Thresholding. Theorem 3.2 provides the formal justification for the hyper-parameters used in the dual-layer mechanism. The convergence neighborhood is determined by the interaction between the two trust layers:

- **Coverage Control:** The hyper-parameter $C_{\max}$ controls the **Reference Evaluation Error**. By forcing a behavior policy update whenever $\mathcal{C} > C_{\max}$, the algorithm ensures the functional critic’s base error $\bar{\epsilon}$ remains manageable, even in complex off-policy settings.
- **Drift Control:** By maintaining $\Delta_{k,t} \leq 2$, the algorithm ensures that the **Gradient Generalization Error** remains on the same order as the reference evaluation error, preventing the $\mathcal{O}(\bar{\epsilon}^2)$ term from being amplified by policy drift.

This result confirms that the policy sequence converges to a stationary point up to a bias dictated by feature quality ($\mathcal{E}_{\text{approx}}$) and the user-defined tolerances for off-policy data quality ($C_{\max}$).

3.3 Efficient Exploration: Functional Critics as a Structural Requirement for Model-Free PSRL

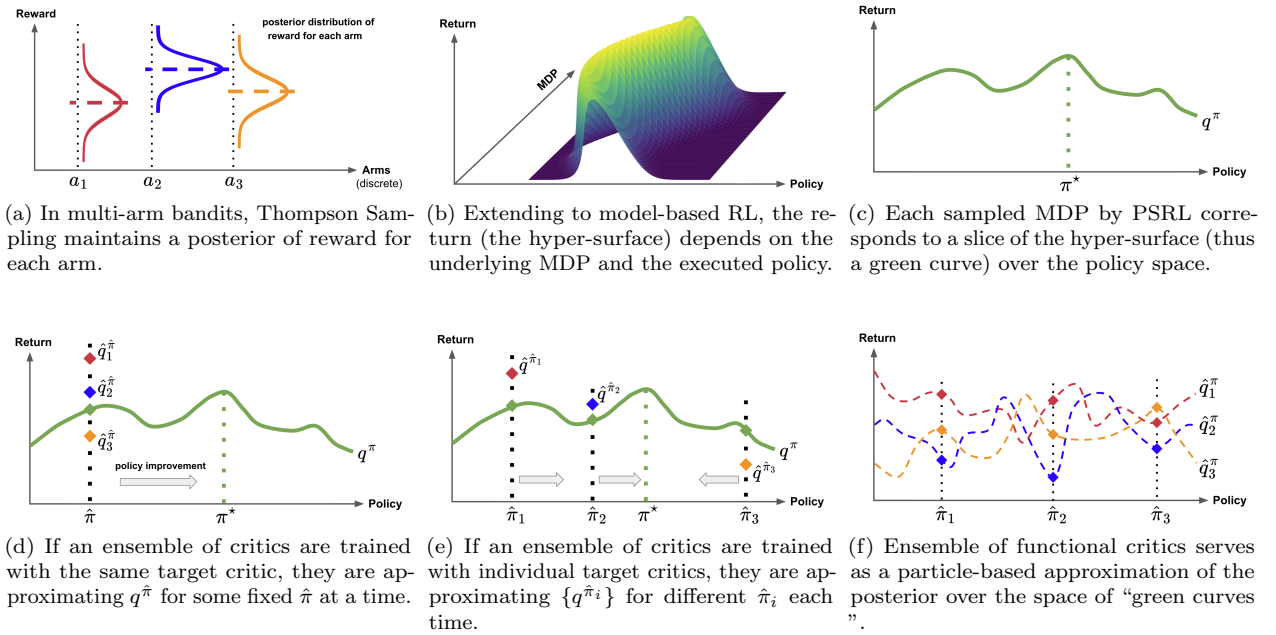


Figure 1: Ensemble of standard critics does not approximate the PSRL while ensemble of functional critics is what we need.

As reviewed in §2, Thompson Sampling (TS) in multi-arm bandits maintains a posterior over each arm’s reward (Figure 1a). In the reinforcement learning setting, however, the expected return is a complex landscape depending on both the underlying MDP and the executed policy (Figure 1b). Posterior Sampling Reinforcement Learning (PSRL) extends the TS principle by maintaining a posterior distribution over MDPs; for each rollout, it samples an MDP from this posterior, effectively “slicing” the return hyper-surface (in Figure 1b) to obtain a policy-conditioned functional graph (green curve in Figure 1c)—the mapping between the policy manifold and the value space. The optimal policy is then extracted from the maximum of this sampled curve and utilized for the rollout.

A faithful model-free approximation of PSRL must therefore maintain a distribution over these return functionals (“green curves”). However, existing approaches (Osband et al., 2016, 2018, 2019) typically utilize ensembles of standard critics, which, by construction, cannot represent a distribution over functionals. This structural limitation becomes evident when examining the two possible methods for constructing target critics for training the ensemble:

1. **Shared Target Critic** (Figure 1d): Utilizing a shared target critic forces all ensemble members to estimate the same quantity $q^{\hat{\pi}}$ for a single, current target policy $\hat{\pi}$. Consequently, the ensemble at best models the uncertainty of a single point on the functional graph. Because it fails to capture the local

geometry or the distribution of the functional itself, it cannot support the policy-dependent (green curve) sampling required for PSRL.

2. **Independent Target Critics** (Figure 1e): Alternatively, using a distinct target critic for each ensemble member results in independent estimates of different quantities $q^{\hat{\pi}_i}$, for different target policies $\hat{\pi}_i$. In this case, the ensemble merely models a sparse set of sampled points on a single curve, rather than any distribution over the curves themselves. In fact, estimates among ensemble members may not even be strictly comparable; thus, variations among them fail to capture uncertainty in any meaningful sense. This approach again lacks the structural capacity to enable true posterior sampling-style exploration.

Ensemble of Functional Critics. We contend that true posterior sampling in a model-free setting necessitates an ensemble of functional critics, where each member represents an independent estimate of the policy-conditioned return functional (Figure 1f). This structural requirement allows the ensemble to serve as a particle-based representation of the posterior over the return functional. By capturing the "entire curve" rather than isolated points, this approach provides a mathematically coherent basis for modeling epistemic uncertainty across the policy manifold. Driven by this insight, we extend Algorithm 1 to employ an ensemble of functional critics paired with corresponding actors, as detailed in Algorithm 4. In this framework, each actor-critic pair evolves as a sample from the joint posterior of policies and their associated return functionals, enabling principled, uncertainty-aware exploration without the need for an explicit MDP model.

Algorithm 4 Functional Actor-Critic Algorithm with Ensemble of Actor-Critic Pairs

- 1: **Inputs:** Initial actor parameter $\{\theta_0^{(i)}\}_{i=1}^n$, initial functional critic parameter $\{\xi_0^{(i)}\}_{i=1}^n$, number of epochs T , batch size m , actor update step-size schedule $\{\eta_t\}_{t=1}^T$
 - 2: **for** $t = 1, \dots, T$ **do**
 - 3: Sample $\mathcal{D}_t \leftarrow (s_t, a_t, r_t, s'_t)$ from the data-collection policy.
 - 4: Update functional critic parameter $\xi_t^{(i)} \leftarrow \text{Functional Policy Evaluator}(t, \xi_{t-1}^{(i)}, \theta_{t-1}^{(i)}, \mathcal{D}_t)$ for each i
 - 5: Compute the **Parameterized Off-Policy Gradient** $G_t^{(i)} \leftarrow \nabla_{\theta}(\mathcal{J}(\pi_{\theta}; \xi_t^{(i)}))|_{\theta=\theta_{t-1}^{(i)}}$ for each i
 - 6: Update the actor parameter $\theta_t^{(i)} \leftarrow \theta_{t-1}^{(i)} - \eta_t G_t^{(i)}$ for each i
 - 7: **end for**
 - 8: **Return** θ_T
-

4 Practical Implementation

To evaluate the utility of functional critics in high-dimensional environments, we extend our framework into a deep reinforcement learning setting. Our implementation leverages the ensemble architecture described in Algorithm 4, utilizing neural networks to parameterize the functional critics, target networks, and actor policies.

Notably, while the explicit dual-layer trust-coverage mechanism of Algorithm 3 is reserved for our theoretical analysis, the practical implementation still inherits the intrinsic stability and generalization benefits of the functional representation. Unlike standard critics that map individual state-action pairs to values, the functional critic explicitly models the value dependency on the structure of the policy manifold. By learning the mapping from policy parameters to the value space, the functional architecture provides a more coherent gradient signal that natively accounts for the policy updates. Furthermore, an ensemble of these functional critics provides posterior-style information over the policy manifold to drive efficient exploration. In this section, we demonstrate how this structural advantage translates to superior performance in both training stability and exploration efficiency in benchmark continuous control tasks.

Functional and Target Critics. We implement an ensemble of functional critics $\{\hat{Q}^{(i)}\}_{i=1}^n$. Each critic comprises three components: a transformer-based actor encoder $E_{act}^{(i)}(\xi_{act}^{(i)})$, a MLP-based state-action encoder $E_{sa}^{(i)}(\xi_{sa}^{(i)})$, and a MLP-based joint encoder $E_{joint}^{(i)}(\xi_{joint}^{(i)})$. The Outputs of $E_{act}^{(i)}$ and $E_{sa}^{(i)}$ are

concatenated and processed by $E_{joint}^{(i)}$ to estimate the state-action value of the input policy:

$$\hat{Q}^{(i)}(\pi_{\theta_t}, s, a; \Xi^{(i)}) = E_{joint}^{(i)}\left(E_{act}^{(i)}(\pi_{\theta_t}; \xi_{act}^{(i)}), E_{sa}^{(i)}(s, a; \xi_{sa}^{(i)}); \xi_{joint}^{(i)}\right),$$

where $\Xi^{(i)} = \{\xi_{act}^{(i)}, \xi_{sa}^{(i)}, \xi_{joint}^{(i)}\}$. For target critics, we maintain delayed copies of E_{sa} and E_{joint} , while sharing the actor encoder E_{act} with the primary functional critic:

$$\hat{Q}_{tar}^{(i)}(\pi_{\theta_t}, s, a; \Xi'^{(i)}) = E_{joint}^{(i)}\left(E_{act}^{(i)}(\pi_{\theta_t}; \xi_{act}^{(i)}), E_{sa}^{(i)}(s, a; \xi_{sa}'^{(i)}); \xi_{joint}'^{(i)}\right),$$

where $\Xi'^{(i)} = \{\xi_{act}^{(i)}, \xi_{sa}'^{(i)}, \xi_{joint}'^{(i)}\}$. Given a target policy π_{θ_t} and a transition (s_t, a_t, r_t, s'_t) , the TD target follows the Bellman equation (3.1):

$$y_t^{(i)} = r_t + \gamma \hat{Q}_{tar}^{(i)}(\pi_{\theta_t}, s'_t, \pi_{\theta_t}(s'_t); \Xi'^{(i)}). \quad (4.1)$$

Each functional critic is trained independently on a product set $\mathcal{A}_t \times \mathcal{D}_t$, where $\mathcal{A}_t = \{\pi_{\theta_t^{(j)}}\}$ is the set of actors and $\mathcal{D}_t$ is a transition batch, by minimizing the squared TD error:

$$L_{\hat{Q}^{(i)}}(\Xi^{(i)}) = (y_t^{(i)} - \hat{Q}^{(i)}(\pi_{\theta_t}, s_t, a_t; \Xi^{(i)}))^2. \quad (4.2)$$

Actor Encoders. The actor encoder E_{act} must balance expressive policy representation with computational efficiency. Drawing from functional analysis, we view the mapping from the policy space Π to $\mathbb{R}$ through evaluation functions. We utilize a set of trainable evaluation states $\{\zeta_i\}_{i=1}^n$ —effectively parameterized delta functions δ_ζ —to probe the input actor, i.e., $\delta_\zeta(\pi) = \pi(\zeta)$. The resulting sequence of output actions and hidden activations from optional layers is processed by a transformer encoder. These evaluation samples $\{\zeta_i\}$ are trained end-to-end as part of the critic parameters $\Xi^{(i)}$ via the loss in (4.2)

Deterministic Actors and Exploration. While most AC methods rely on stochastic actors and entropy tuning, our functional critic’s ability to provide gradients for any actor allows us to employ an ensemble of simple deterministic actors $\{\pi_{\theta_t^{(i)}}\}$. This ensemble serves as the training set $\mathcal{A}_t$ for the critics. To achieve posterior sampling-style exploration, we maintain a fixed pairing between each actor $\pi_{\theta_t^{(i)}}$ and its corresponding critic $\hat{Q}^{(i)}$. Each actor is updated using the exact off-policy gradient (2.2) derived from its paired functional critic.

Omitted Implementation Heuristics. Notably, our framework discards several heuristics currently viewed as essential for off-policy AC. For the critics, we do not employ “clipped double-Q” or “minimum target” value methods (Fujimoto et al., 2018; Haarnoja et al., 2018) to mitigate overestimation. For the actors, we employ deterministic policies without exploration noise or target policy smoothing (Fujimoto et al., 2018), while also omitting the entropy regularization (Haarnoja et al., 2018) typically required for stochastic policies. The omission of these techniques demonstrates that functional critic modeling inherently addresses training stability and exploration without requiring secondary implementation “hacks”.

Algorithm 5 summarizes this minimalist implementation, which translates the theoretical insights of §3 into a practical, high-performance algorithm.

4.1 Empirical results

Baselines and Codebase. We compare our method against SAC (Haarnoja et al., 2018) and TD3 (Fujimoto et al., 2018), the two most widely-adopted off-policy AC baselines. Over years of refinement, the RL community has converged on a high-performance “update recipe” characterized by a high update-to-data (UTD) ratio stabilized by critic ensembles and layer normalization (Chen et al., 2021; Li et al., 2023; Ball et al., 2023). This configuration underpins current state-of-the-art results across both offline and online RL (Zhou et al., 2024; Xiao et al., 2025; Nauman et al., 2024; Seo et al., 2025). We utilize implementations of SAC/TD3 with this update recipe from a rigorously tested open-sourced RL library (Xu et al., 2021), further fine-tuning them to create the enhanced baselines denoted as **SAC+** and **TD3+**. Comparison results in Figure 3 confirm that SAC+ significantly outperforms the original SAC (v2), ensuring our benchmarks reflect modern state-of-the-art performance.

Algorithm 5 Neural Network-based Functional Actor-Critic Algorithm

```
1: Initialize the actor ensemble  $\mathcal{A}_0$  parameters  $\{\theta_0^{(i)}\}_{i=1}^n$ 
2: initialize functional critic ensemble parameters  $\{\xi_{act}^{(i)}, \xi_{sa}^{(i)}, \xi_{joint}^{(i)}\}_{i=1}^n$ 
3: Initialize extra target functional critic ensemble parameters  $\{\xi_{sa}'^{(i)}, \xi_{joint}'^{(i)}\}_{i=1}^n$ 
4: Initialize empty replay buffer  $\mathcal{R}$ 
5: Select number of epochs  $T$ , transition batch size  $m$  for training, functional critic UTD  $G$ 
6: for  $t = 1, \dots$  do
7:   Resample rollout actor index id from  $\{1, \dots, n\}$  if starting a new episode
8:   Take action  $a_t = \pi_{\theta^{(id)}}(s_t)$ 
9:   Store transition  $(s_t, a_t, r_t, s_{t+1})$  to buffer  $\mathcal{R}$ 
10:  for  $g = 1, \dots, G$  do
11:    Sample transition batch  $B_C$  from the buffer  $\mathcal{R}$ 
12:    for  $i = 1, \dots, n$  do
13:      Compute TD targets (4.1) for batch  $\mathcal{A}_t \times B_C$ 
14:      Update  $\{\xi_{act}^{(i)}, \xi_{sa}^{(i)}, \xi_{joint}^{(i)}\}$  minimizing a batched  $(\mathcal{A}_t \times B_C)$  version of (4.2)
15:    end for
16:    Update target critics  $\xi_{sa}'^{(i)} \leftarrow \rho \xi_{sa}'^{(i)} + (1 - \rho) \xi_{sa}^{(i)}$ ,  $\xi_{joint}'^{(i)} \leftarrow \rho \xi_{joint}'^{(i)} + (1 - \rho) \xi_{joint}^{(i)}$ 
17:  end for
18:  Sample transition batch  $B_A$  from the buffer  $\mathcal{R}$ 
19:  for  $i = 1, \dots, n$  do
20:    Evaluate exact off-policy gradient (2.2) with  $\hat{Q}^{(i)}$  and  $B_A$ 
21:    Update  $\theta_t^{(i)}$ 
22:  end for
23: end for
```

Existing Functional Critic Methods. We exclude prior functional critic (or Policy-Conditioned Value Function, PCVF) methods from this comparison. A comprehensive evaluation (Griesbach, 2023) found that none of the major PCVF variants were competitive with standard RL algorithms. This lack of performance is consistent with our observation that prior works lacked the critical insights regarding training stability and exploration identified in this paper. We refer readers to Griesbach (2023) for a detailed benchmarking of those earlier approaches.

Experimental Setup and Fairness. To ensure a controlled comparison, we implemented our algorithm within the same codebase as the baselines. We carefully aligned all shared hyperparameters: both our method and the SAC+/TD3+ baselines employ 10 critics (and target critics), and all networks share a common backbone architecture where possible. Deviations are strictly method-specific; for instance, our functional critic includes an actor-encoder module to process policy inputs, while the SAC+ actor incorporates Gaussian projection layers for stochasticity and TD3 utilizes exploration noise for its deterministic policy during rollouts. Detailed hyperparameters and architectural configurations are provided in Appendix A.

Results and Analysis. We evaluated our approach alongside the SAC+ and TD3+ baselines on the Cheetah-run and Hopper-hop continuous control tasks of the DeepMind Control Suite (Tunyasuvunakool et al., 2020). The results are reported in Figure 2. Despite the preliminary nature of these experiments, we observe several encouraging indicators of our framework’s effectiveness.

First, our approach achieves performance favorable to representative state-of-the-art method without relying on widely adopted heuristics typically considered essential for off-policy AC stability. Second, while SAC+ requires a high update-to-data (UTD) ratio (In our experiments, 10 critic updates per actor update after grid search), our approach performs robustly with a critic-to-actor update ratio of just 2 or 3. We fix this ratio at 3 across all experiments. These findings validate our theoretical insights that functional critic modeling stabilizes critic training without necessitating a rigid multi-timescale actor-critic design.

Notably, we have not yet achieved stable training at a 1:1 actor-critic update ratio. We attribute this to our relatively simple actor encoder design, which only extracts neurons from the final two layers of the actor

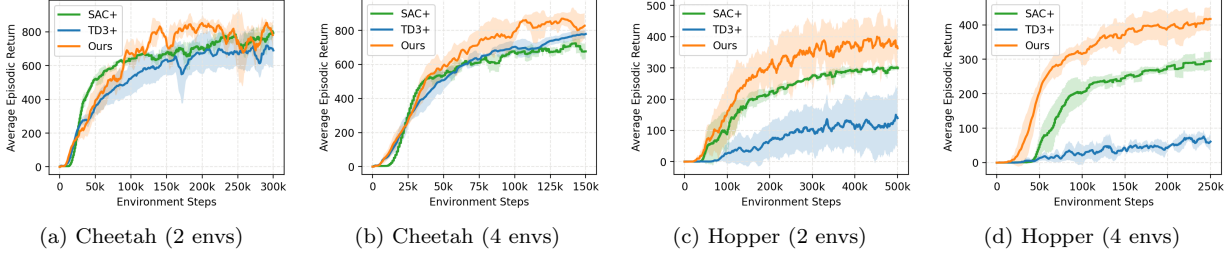


Figure 2: Averaged episodic return against environment steps of our method vs SAC+ and TD3+ on Cheetah-run and Hopper-hop tasks of DM Control. “n envs” means the number of parallel environments. Results are averaged over four runs of different random seeds, with the shaded area corresponding to the standard deviation.

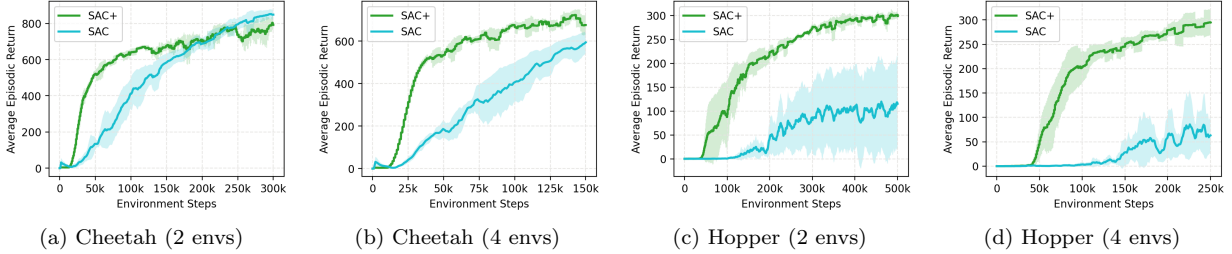


Figure 3: Averaged episodic return against environment steps of SAC+ vs SAC on Cheetah-run and Hopper-hop tasks of DM Control. “n envs” means the number of parallel environments. Results are averaged over four runs of different random seeds, with the shaded area corresponding to the standard deviation.

network to form the input sequence—a choice made mainly for efficiency. Consequently, the full generalization potential of the functional critic in the input actor space has not yet been realized. Finally, our method better leverages the parallel environments: the performance gap over SAC+/TD3+ consistently widens as the number of environments increases from two to four. We attribute this to the data-driven generalization inherent in functional critic modeling, suggesting even greater potential if combined with more sophisticated exploration strategies and more aggressive parallelization.

5 Conclusion and Future Work

In this work, we revisit functional critics within the off-policy actor-critic framework, uncovering their fundamental significance in ensuring training stability, efficient policy improvement, and principled exploration. By leveraging the structural benefits of linear functional representation, we establish the first convergence guarantees for off-policy AC that utilize target-based TD, accommodate evolving behavior policies, and replace restrictive “full coverage” assumptions with a data-driven dual-layer trust-coverage mechanism. These results dismantle longstanding barriers between theoretical stability and practical RL recipes. Furthermore, our analysis reveals a deep link between functional critics and posterior sampling-style exploration, providing a new perspective on the intersection of function approximation and uncertainty-aware RL.

Moving forward, several promising directions remain:

- **Trust-Criteria Implementation:** The trust coverage criteria in our theoretical Algorithm 3 utilize the linear functional representation and its associated covariance matrices. While our minimalist implementation demonstrates robustness without them, developing a sample-based, neural network-compatible estimator for $\mathcal{C}(\pi, \mu)$ and $\Delta_{k,t}$ could further stabilize deep RL training and improve sample efficiency in highly non-stationary environments.
- **Adaptive Behavior Control:** Our analysis suggests that behavior policies should be optimized to maximize the condition of the feature covariance $\mathbf{A}_k$. Actively adapting μ to minimize the evaluation trust metric $\mathcal{C}(\pi, \mu)$ presents a promising path toward superior sample efficiency in off-policy setting.

- **Trust-Region Optimization:** Integrating the gradient generalization metric $\Delta_{k,t}$ directly into the actor’s objective as a functional constraint—mirroring the success of KL-divergence constraints in TRPO and PPO—could provide a more robust, data-aware mechanism for policy improvement. This would allow the actor to automatically slow its update rate when entering feature regimes where the functional critic’s predictions are less reliable.

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A Implementation Details

Hyperparameters across all experiments reported in Figure 2 is summarize in Table 1.

Table 1: Hyperparameters used across all reported experiments.

Hyperparameter	Value
optimizer	Adam
learning rate	3×10^{-4}
batch size	256
discount factor (γ)	0.99
target update rate (τ)	0.005
actor network hidden layers	(256, 256)
critic state-action encoder hidden layers	(256, 256)
critic joint encoder hidden layers	(256, 256)
actor transformer encoder number of layers	4
actor transformer encoder number of heads	1
size of trainable evaluation samples for actor encoder	512
actor network layers to extract hidden neurons for actor encoder	last two
actor encoding dimension	128
state-action encoding dimension	64
activation function	ReLU

B Proof of Results in Section 3.2

B.1 Geometric Perspective on Assumption 2

In this section, we provide a geometric perspective for understanding the linear functional representation of Assumption 2. We consider the policy space Π as a smooth Riemannian manifold. For any fixed state-action pair (s, a) , the value function can be understood as a functional map $\mathcal{Q}_{s,a} : \Pi \rightarrow \mathbb{R}$. While standard linear function approximation focuses on the representability of a single policy, Assumption 2 leverages the intrinsic local continuity and smoothness of the policy manifold Π .

For any anchored reference policy $\pi_0 \in \Pi$, let $T_{\pi_0}\Pi$ denote the tangent space at π_0 , with the associated exponential map $\exp_{\pi_0}(\cdot) : T_{\pi_0}\Pi \rightarrow \Pi$ and its inverse, the logarithmic map $\log_{\pi_0}(\cdot) : \Pi \rightarrow T_{\pi_0}\Pi$. For any policy π in a sufficiently small neighborhood of π_0 , a first-order Taylor expansion on the manifold yields:

$$Q^\pi(s, a) = \mathcal{Q}_{s,a}(\exp_{\pi_0}(\log_{\pi_0}(\pi))) \approx Q^{\pi_0}(s, a) + \langle \nabla_\pi \mathcal{Q}_{s,a}(\pi_0), \log_{\pi_0}(\pi) \rangle_{T_{\pi_0}\Pi},$$

where $\nabla_\pi \mathcal{Q}_{s,a}(\pi_0)$ represents the functional gradient of the value with respect to the anchor policy.

By identifying a set of basis functions $\Phi(s, a)$ that span the directional derivatives of the value functional, the displacement vector $\log_{\pi_0}(\pi) \in T_{\pi_0}\Pi$ interacts linearly with the state-action features. Specifically, we can define the **functional feature map** as the tensor product $\phi^\pi(s, a) := \text{vec}(\Phi(s, a) \otimes \log_{\pi_0}(\pi))$. Under this coordinate-free construction, the value function locally satisfies:

$$Q^\pi(s, a) \approx \phi^\pi(s, a)^\top w_{\pi_0}.$$

In this light, Assumption 2 holds locally for any π close to π_0 , with the underlying representation depending on the choice of the anchor policy. Our dual-layer mechanism is precisely designed to respect the boundaries of this local validity: while the first layer ($\mathcal{C}(\pi, \mu)$) ensures the data is sufficient to build the anchored critic, the second layer ($\Delta_{k,t}$) monitors the displacement on the tangent space. Crucially, the **Gradient Generalization Trust Metric** $\Delta_{k,t}$ serves as a data-aware proxy for the Riemannian distance $d(\pi, \pi_0)$ on the manifold. By enforcing $\Delta_{k,t} \leq 2$, Algorithm 3 refreshes the anchor policy exactly when the policy π drifts beyond the "trust region" where the manifold's first-order approximation—and thus the functional critic's gradient—remains reliable.

B.2 A Finite-time Bound for Markovian SAA under (Possibly) Degenerate Norm

In this section, we provide a detailed proof of a finite-time contraction bound for Markovian stochastic approximation under a possibly degenerate design, which serves as a foundation for our proof of Theorem 3.1 in the next section. The proof follows a Lyapunov drift argument as in [Chen et al. \(2022\)](#) while carefully tracking the semi-norm induced by a positive semi-definite matrix.

Notations. For any positive semi-definite matrix $\mathbf{A} \succeq \mathbf{0}$ and vectors $\mathbf{u}, \mathbf{v}$ with the same dimension, we define $\langle \mathbf{u}, \mathbf{v} \rangle_{\mathbf{A}} := \mathbf{u}^\top \mathbf{A} \mathbf{v}$ and $\|\mathbf{v}\|_{\mathbf{A}}^2 := \mathbf{v}^\top \mathbf{A} \mathbf{v}$. Let $\lambda_{\max}(\mathbf{A})$ denote the largest eigenvalue of $\mathbf{A}$. When $\mathbf{A}$ is singular, let $\lambda_{\min}^+(\mathbf{A})$ denote the smallest *strictly positive* eigenvalue of $\mathbf{A}$ (and W.L.O.G. $\lambda_{\min}^+(\mathbf{A}) = 0$ when $\mathbf{A} = \mathbf{0}$). Let $\Pi_{\mathbf{A}}$ be the orthogonal projection operator onto $\text{Range}(\mathbf{A})$ and $\Pi_{\mathbf{A}}^\perp$ the orthogonal projection operator onto $\ker(\mathbf{A})$, so that $\Pi_{\mathbf{A}} + \Pi_{\mathbf{A}}^\perp = \mathbf{I}$. For a finite-state Markov chain $\{\mathbf{y}_k\}_{k \geq 0}$ over $\mathcal{Y}$ with transition kernel P and stationary distribution ν , we define the β -mixing time as

$$\tau_\beta := \min \{k \geq 0 : \max_{\mathbf{y} \in \mathcal{Y}} \|P^k(\mathbf{y}, \cdot) - \nu(\cdot)\|_{\text{TV}} \leq \beta\}.$$

Theorem B.1. Consider the Markovian stochastic approximation recursion with arbitrary initialization $\mathbf{x}_0 \in \mathbb{R}^d$ and

$$\mathbf{x}_{k+1} = \mathbf{x}_k + \alpha \mathbf{F}(\mathbf{x}_k, \mathbf{y}_k). \quad (\text{B.1})$$

Let $\bar{\mathbf{F}}(\mathbf{x}) := \mathbb{E}_{\mathbf{y} \sim \nu}[\mathbf{F}(\mathbf{x}, \mathbf{y})]$. Suppose that for some $\mathbf{A} \succeq \mathbf{0}, \mathbf{A} \neq \mathbf{0}$:

1. (**Geometric mixing**). The chain $\{\mathbf{y}_k\}$ has a unique stationary distribution ν , and $\max_{\mathbf{y} \in \mathcal{Y}} \|P^k(\mathbf{y}, \cdot) - \nu(\cdot)\|_{\text{TV}} \leq C\rho^k$ for some $C > 0$ and $\rho \in (0, 1)$.
2. (**Euclidean Lipschitz + boundedness**). There exist $L_1, L_2 > 0$ such that for all $\mathbf{x}_1, \mathbf{x}_2 \in \mathbb{R}^d$ and $\mathbf{y} \in \mathcal{Y}$,

$$\|\mathbf{F}(\mathbf{x}_1, \mathbf{y}) - \mathbf{F}(\mathbf{x}_2, \mathbf{y})\|_2 \leq L_1 \|\mathbf{x}_1 - \mathbf{x}_2\|_2, \quad \|\mathbf{F}(\mathbf{0}, \mathbf{y})\|_2 \leq L_2.$$

3. (**Semi-norm drift**). The equation $\bar{\mathbf{F}}(\mathbf{x}) = \mathbf{0}$ has at least one solution (not necessarily unique), and fix any such $\mathbf{x}^*$. Moreover, for all $\mathbf{x} \in \mathbb{R}^d$,

$$\langle \mathbf{x} - \mathbf{x}^*, \bar{\mathbf{F}}(\mathbf{x}) \rangle_{\mathbf{A}} \leq -\kappa \|\mathbf{x} - \mathbf{x}^*\|_{\mathbf{A}}^2 \quad (\text{B.2})$$

for some constant $\kappa > 0$.

4. (**Range-compatibility for degenerate $\mathbf{A}$**). For all $\mathbf{x} \in \mathbb{R}^d$ and $\mathbf{y} \in \mathcal{Y}$, the increment satisfies

$$\mathbf{F}(\mathbf{x}, \mathbf{y}) \in \text{Range}(\mathbf{A}) \quad \text{equivalently} \quad \Pi_{\mathbf{A}}^\perp \mathbf{F}(\mathbf{x}, \mathbf{y}) = \mathbf{0}. \quad (\text{B.3})$$

Define $L := \max\{L_1, L_2\}$. Assume $\mathbf{A} \neq \mathbf{0}$ so that $\lambda_{\min}^+(\mathbf{A})$ is well-defined, and choose α such that

$$\alpha \tau_\alpha \leq \min \left\{ \frac{1}{4L}, \frac{\kappa \lambda_{\min}^+(\mathbf{A})}{260 \lambda_{\max}(\mathbf{A}) L^2} \right\}. \quad (\text{B.4})$$

Then for all $k \geq \tau_\alpha$,

$$\mathbb{E}[\|\mathbf{x}_{k+1} - \mathbf{x}^*\|_{\mathbf{A}}^2] \leq (1 - \kappa\alpha) \mathbb{E}[\|\mathbf{x}_k - \mathbf{x}^*\|_{\mathbf{A}}^2] + \frac{260 \lambda_{\max}(\mathbf{A}) L^2}{\lambda_{\min}^+(\mathbf{A})} \alpha^2 \tau_\alpha \cdot \mathcal{C}_*^2, \quad (\text{B.5})$$

where

$$\mathcal{C}_*^2 := \|\Pi_{\mathbf{A}}^\perp(\mathbf{x}_0 - \mathbf{x}^*)\|_2^2 + (\|\mathbf{x}^*\|_2 + 1)^2. \quad (\text{B.6})$$

Proof of Theorem B.1. Fix $t = \tau_\alpha$ and define the conditional expectation

$$\mathbb{E}_k[\cdot] := \mathbb{E}[\cdot \mid \mathbf{x}_{k-t}, \mathbf{y}_{k-t}], \quad k \geq t.$$

We begin with a one-step expansion of the $\mathbf{A}$ -semi-norm:

$$\begin{aligned} \mathbb{E}_k[\|\mathbf{x}_{k+1} - \mathbf{x}^*\|_{\mathbf{A}}^2] - \mathbb{E}_k[\|\mathbf{x}_k - \mathbf{x}^*\|_{\mathbf{A}}^2] &= 2\mathbb{E}_k[\langle \mathbf{x}_k - \mathbf{x}^*, \mathbf{x}_{k+1} - \mathbf{x}_k \rangle_{\mathbf{A}}] + \mathbb{E}_k[\|\mathbf{x}_{k+1} - \mathbf{x}_k\|_{\mathbf{A}}^2] \\ &= \underbrace{2\alpha \mathbb{E}_k[\langle \mathbf{x}_k - \mathbf{x}^*, \bar{\mathbf{F}}(\mathbf{x}_k) \rangle_{\mathbf{A}}]}_{:= (\text{a})} + \underbrace{2\alpha \mathbb{E}_k[\langle \mathbf{x}_k - \mathbf{x}^*, \mathbf{F}(\mathbf{x}_k, \mathbf{y}_k) - \bar{\mathbf{F}}(\mathbf{x}_k) \rangle_{\mathbf{A}}]}_{:= (\text{b})} + \underbrace{\alpha^2 \mathbb{E}_k[\|\mathbf{F}(\mathbf{x}_k, \mathbf{y}_k)\|_{\mathbf{A}}^2]}_{:= (\text{c})}. \end{aligned}$$

Now it remains to bound three terms (a)–(c) separately.

Step 1: negative drift term (a). We have it holds by (B.2) that

$$2\alpha \mathbb{E}_k [\langle \mathbf{x}_k - \mathbf{x}^*, \bar{\mathbf{F}}(\mathbf{x}_k) \rangle_{\mathbf{A}}] \leq -2\kappa\alpha \mathbb{E}_k [\|\mathbf{x}_k - \mathbf{x}^*\|_{\mathbf{A}}^2]. \quad (\text{B.7})$$

To control terms (b), (c), we first introduce several preliminary lemmas as the following:

Lemma 1 (Global growth bound). *Under Assumption 2, for all $\mathbf{x} \in \mathbb{R}^d$ and $\mathbf{y} \in \mathcal{Y}$,*

$$\|\mathbf{F}(\mathbf{x}, \mathbf{y})\|_2 \leq L(\|\mathbf{x}\|_2 + 1), \quad \|\mathbf{F}(\mathbf{x}, \mathbf{y})\|_{\mathbf{A}} \leq \sqrt{\lambda_{\max}(\mathbf{A})} L(\|\mathbf{x}\|_2 + 1).$$

The same bounds hold for $\|\bar{\mathbf{F}}(\mathbf{x})\|_2$ and $\|\bar{\mathbf{F}}(\mathbf{x})\|_{\mathbf{A}}$.

Proof. By Lipschitzness and the bound at the origin,

$$\|\mathbf{F}(\mathbf{x}, \mathbf{y})\|_2 \leq \|\mathbf{F}(\mathbf{x}, \mathbf{y}) - \mathbf{F}(\mathbf{0}, \mathbf{y})\|_2 + \|\mathbf{F}(\mathbf{0}, \mathbf{y})\|_2 \leq L_1 \|\mathbf{x}\|_2 + L_2 \leq L(\|\mathbf{x}\|_2 + 1).$$

For the $\mathbf{A}$ -semi-norm, use $\|\mathbf{v}\|_{\mathbf{A}}^2 = \mathbf{v}^\top \mathbf{A} \mathbf{v} \leq \lambda_{\max}(\mathbf{A}) \|\mathbf{v}\|_2^2$. Finally, $\bar{\mathbf{F}}(\mathbf{x}) = \mathbb{E}_{\mathbf{y} \sim \nu} [\mathbf{F}(\mathbf{x}, \mathbf{y})]$ and Jensen's inequality give the same bounds for $\bar{\mathbf{F}}$. $\square$

Lemma 2 (Range invariance and change-of-norm bounds). *Assume (B.3). Then for all $k \geq 0$,*

$$\Pi_{\mathbf{A}}^\perp \mathbf{x}_k = \Pi_{\mathbf{A}}^\perp \mathbf{x}_0, \quad \Pi_{\mathbf{A}}^\perp (\mathbf{x}_k - \mathbf{x}^*) = \Pi_{\mathbf{A}}^\perp (\mathbf{x}_0 - \mathbf{x}^*).$$

Moreover, for any $\mathbf{v} \in \mathbb{R}^d$,

$$\|\Pi_{\mathbf{A}} \mathbf{v}\|_2^2 \leq \frac{1}{\lambda_{\min}^+(\mathbf{A})} \|\mathbf{v}\|_{\mathbf{A}}^2, \quad \|\mathbf{v}\|_2^2 \leq \frac{1}{\lambda_{\min}^+(\mathbf{A})} \|\mathbf{v}\|_{\mathbf{A}}^2 + \|\Pi_{\mathbf{A}}^\perp \mathbf{v}\|_2^2. \quad (\text{B.8})$$

Proof. By (B.3), we have

$$\Pi_{\mathbf{A}}^\perp \mathbf{x}_{k+1} = \Pi_{\mathbf{A}}^\perp \mathbf{x}_k + \alpha \Pi_{\mathbf{A}}^\perp \mathbf{F}(\mathbf{x}_k, \mathbf{y}_k) = \Pi_{\mathbf{A}}^\perp \mathbf{x}_k,$$

hence $\Pi_{\mathbf{A}}^\perp \mathbf{x}_k = \Pi_{\mathbf{A}}^\perp \mathbf{x}_0$ for all k . The identity for $\mathbf{x}_k - \mathbf{x}^*$ follows immediately.

For (B.8), write the eigendecomposition $\mathbf{A} = \mathbf{U} \text{diag}(\lambda_1, \dots, \lambda_d) \mathbf{U}^\top$ with $\lambda_i \geq 0$. Let $\mathbf{U}_+$ collect eigenvectors with $\lambda_i > 0$. Then $\Pi_{\mathbf{A}} = \mathbf{U}_+ \mathbf{U}_+^\top$ and

$$\|\mathbf{v}\|_{\mathbf{A}}^2 = \sum_{\lambda_i > 0} \lambda_i (\mathbf{u}_i^\top \mathbf{v})^2 \geq \lambda_{\min}^+(\mathbf{A}) \sum_{\lambda_i > 0} (\mathbf{u}_i^\top \mathbf{v})^2 = \lambda_{\min}^+(\mathbf{A}) \|\Pi_{\mathbf{A}} \mathbf{v}\|_2^2,$$

which gives the first inequality. The second follows from $\|\mathbf{v}\|_2^2 = \|\Pi_{\mathbf{A}} \mathbf{v}\|_2^2 + \|\Pi_{\mathbf{A}}^\perp \mathbf{v}\|_2^2$. $\square$

Lemma 3 (Mixing bias for Markovian sampling). *For any $k \geq t = \tau_\alpha$ and any deterministic $\mathbf{x} \in \mathbb{R}^d$,*

$$\left\| \mathbb{E}[\mathbf{F}(\mathbf{x}, \mathbf{y}_k) \mid \mathbf{y}_{k-t} = y] - \bar{\mathbf{F}}(\mathbf{x}) \right\|_2 \leq 2L\alpha (\|\mathbf{x}\|_2 + 1) \quad \text{for all } y \in \mathcal{Y}.$$

Consequently,

$$\left\| \mathbb{E}[\mathbf{F}(\mathbf{x}, \mathbf{y}_k) \mid \mathbf{y}_{k-t} = y] - \bar{\mathbf{F}}(\mathbf{x}) \right\|_{\mathbf{A}} \leq 2\sqrt{\lambda_{\max}(\mathbf{A})} L\alpha (\|\mathbf{x}\|_2 + 1).$$

Proof. Since $\mathcal{Y}$ is finite, we can write

$$\mathbb{E}[\mathbf{F}(\mathbf{x}, \mathbf{y}_k) \mid \mathbf{y}_{k-t} = y] = \sum_{y' \in \mathcal{Y}} P^t(y, y') \mathbf{F}(\mathbf{x}, y'), \quad \bar{\mathbf{F}}(\mathbf{x}) = \sum_{y' \in \mathcal{Y}} \nu(y') \mathbf{F}(\mathbf{x}, y').$$

Hence,

$$\begin{aligned} \left\| \mathbb{E}[\mathbf{F}(\mathbf{x}, \mathbf{y}_k) \mid \mathbf{y}_{k-t} = y] - \bar{\mathbf{F}}(\mathbf{x}) \right\|_2 &\leq \sum_{y' \in \mathcal{Y}} |P^t(y, y') - \nu(y')| \|\mathbf{F}(\mathbf{x}, y')\|_2 \\ &\leq 2\|P^t(y, \cdot) - \nu(\cdot)\|_{\text{TV}} \cdot \max_{y' \in \mathcal{Y}} \|\mathbf{F}(\mathbf{x}, y')\|_2 \\ &\leq 2\alpha \cdot L(\|\mathbf{x}\|_2 + 1), \end{aligned}$$

where we used the definition of total variation and Lemma 1 for the last step. The $\mathbf{A}$ -bound follows from $\|\cdot\|_{\mathbf{A}} \leq \sqrt{\lambda_{\max}(\mathbf{A})} \|\cdot\|_2$. $\square$

Finally, we show the following lemma, which mirrors [Chen et al. \(2022\)](#) (Lemma 2.3 therein) for constant stepsize.

Lemma 4. Assume $\alpha t \leq \frac{1}{4L}$. Then for any $k \geq t$,

$$\|\mathbf{x}_k - \mathbf{x}_{k-t}\|_2 \leq 2L\alpha t (\|\mathbf{x}_{k-t}\|_2 + 1), \quad (\text{B.9})$$

$$\|\mathbf{x}_k - \mathbf{x}_{k-t}\|_2 \leq 4L\alpha t (\|\mathbf{x}_k\|_2 + 1). \quad (\text{B.10})$$

Consequently,

$$\|\mathbf{x}_k - \mathbf{x}_{k-t}\|_{\mathbf{A}} \leq \sqrt{\lambda_{\max}(\mathbf{A})} \|\mathbf{x}_k - \mathbf{x}_{k-t}\|_2 \leq 2\sqrt{\lambda_{\max}(\mathbf{A})} L\alpha t (\|\mathbf{x}_{k-t}\|_2 + 1).$$

Proof. From the recursion,

$$\|\mathbf{x}_{j+1}\|_2 - \|\mathbf{x}_j\|_2 \leq \|\mathbf{x}_{j+1} - \mathbf{x}_j\|_2 = \alpha \|\mathbf{F}(\mathbf{x}_j, \mathbf{y}_j)\|_2 \leq \alpha L(\|\mathbf{x}_j\|_2 + 1),$$

so $(\|\mathbf{x}_{j+1}\|_2 + 1) \leq (1 + L\alpha)(\|\mathbf{x}_j\|_2 + 1)$. Iterating from $j = k - t$ to $k - 1$ yields

$$\|\mathbf{x}_j\|_2 + 1 \leq (1 + L\alpha)^{j-(k-t)} (\|\mathbf{x}_{k-t}\|_2 + 1) \leq e^{L\alpha t} (\|\mathbf{x}_{k-t}\|_2 + 1).$$

Since $\alpha t \leq \frac{1}{4L}$, we have $L\alpha t \leq \frac{1}{4}$ and thus $e^{L\alpha t} \leq 1 + 2L\alpha t \leq 2$. Therefore $\|\mathbf{x}_j\|_2 + 1 \leq 2(\|\mathbf{x}_{k-t}\|_2 + 1)$ for all $j \in [k - t, k]$. Now sum increments:

$$\|\mathbf{x}_k - \mathbf{x}_{k-t}\|_2 \leq \sum_{j=k-t}^{k-1} \|\mathbf{x}_{j+1} - \mathbf{x}_j\|_2 = \alpha \sum_{j=k-t}^{k-1} \|\mathbf{F}(\mathbf{x}_j, \mathbf{y}_j)\|_2 \leq \alpha \sum_{j=k-t}^{k-1} L(\|\mathbf{x}_j\|_2 + 1) \leq \alpha t \cdot 2L(\|\mathbf{x}_{k-t}\|_2 + 1),$$

which is (B.9). The bound (B.10) follows similarly by running the same argument backward (or using $\|\mathbf{x}_{k-t}\|_2 + 1 \leq 2(\|\mathbf{x}_k\|_2 + 1)$, which follows from the same estimate with indices swapped). $\square$

Step 2: Bounding term (b) Inspired by Lemma 2.4 of [Chen et al. \(2022\)](#) we establish the following result based on above lemmas regarding term (b)

Lemma 5 (Markovian noise inner product bound). Assume $\alpha t \leq \frac{1}{4L}$. Then for any $k \geq t$,

$$\mathbb{E}_k [\langle \mathbf{x}_k - \mathbf{x}^*, \mathbf{F}(\mathbf{x}_k, \mathbf{y}_k) - \bar{\mathbf{F}}(\mathbf{x}_k) \rangle_{\mathbf{A}}] \leq \frac{64 \lambda_{\max}(\mathbf{A}) L^2}{\lambda_{\min}^+(\mathbf{A})} \alpha t \left(\mathbb{E}_k [\|\mathbf{x}_k - \mathbf{x}^*\|_{\mathbf{A}}^2] + \lambda_{\max}(\mathbf{A}) \mathcal{C}_{\star}^2 \right), \quad (\text{B.11})$$

where $\mathcal{C}_{\star}$ is defined in (B.6).

Proof. Write $t = \tau_{\alpha}$ and decompose, for $k \geq t$:

$$\begin{aligned} & \mathbb{E}_k [\langle \mathbf{x}_k - \mathbf{x}^*, \mathbf{F}(\mathbf{x}_k, \mathbf{y}_k) - \bar{\mathbf{F}}(\mathbf{x}_k) \rangle_{\mathbf{A}}] \\ &= \mathbb{E}_k [\langle \mathbf{x}_k - \mathbf{x}_{k-t}, \mathbf{F}(\mathbf{x}_k, \mathbf{y}_k) - \bar{\mathbf{F}}(\mathbf{x}_k) \rangle_{\mathbf{A}}] \\ & \quad + \langle \mathbf{x}_{k-t} - \mathbf{x}^*, \mathbb{E}_k [\mathbf{F}(\mathbf{x}_{k-t}, \mathbf{y}_k)] - \bar{\mathbf{F}}(\mathbf{x}_{k-t}) \rangle_{\mathbf{A}} \\ & \quad + \mathbb{E}_k [\langle \mathbf{x}_{k-t} - \mathbf{x}^*, \mathbf{F}(\mathbf{x}_k, \mathbf{y}_k) - \mathbf{F}(\mathbf{x}_{k-t}, \mathbf{y}_k) \rangle_{\mathbf{A}}] \\ & \quad + \mathbb{E}_k [\langle \mathbf{x}_{k-t} - \mathbf{x}^*, \bar{\mathbf{F}}(\mathbf{x}_{k-t}) - \bar{\mathbf{F}}(\mathbf{x}_k) \rangle_{\mathbf{A}}]. \end{aligned}$$

We bound the four terms.

Bound for (T1). By Cauchy-Schwarz in the $\mathbf{A}$ -inner product and Lemma 1,

$$\|\mathbf{F}(\mathbf{x}_k, \mathbf{y}_k) - \bar{\mathbf{F}}(\mathbf{x}_k)\|_{\mathbf{A}} \leq \|\mathbf{F}(\mathbf{x}_k, \mathbf{y}_k)\|_{\mathbf{A}} + \|\bar{\mathbf{F}}(\mathbf{x}_k)\|_{\mathbf{A}} \leq 2\sqrt{\lambda_{\max}(\mathbf{A})} L(\|\mathbf{x}_k\|_2 + 1).$$

Also, Lemma 4 implies

$$\|\mathbf{x}_k - \mathbf{x}_{k-t}\|_{\mathbf{A}} \leq \sqrt{\lambda_{\max}(\mathbf{A})} \|\mathbf{x}_k - \mathbf{x}_{k-t}\|_2 \leq 4\sqrt{\lambda_{\max}(\mathbf{A})} L\alpha t (\|\mathbf{x}_k\|_2 + 1),$$

where we used (B.10). Therefore,

$$(\text{T1}) \leq 8 \lambda_{\max}(\mathbf{A}) L^2 \alpha t \mathbb{E}_k [(\|\mathbf{x}_k\|_2 + 1)^2]. \quad (\text{B.16})$$

Bound for (T2). By Cauchy–Schwarz and Lemma 3,

$$(T2) \leq \mathbb{E}_k [\|\mathbf{x}_{k-t} - \mathbf{x}^*\|_{\mathbf{A}}] \cdot 2\sqrt{\lambda_{\max}(\mathbf{A})} L\alpha (\|\mathbf{x}_{k-t}\|_2 + 1).$$

Using $ab \leq \frac{1}{2}(a^2 + b^2)$ and then Lemma 4 (which implies $\|\mathbf{x}_{k-t}\|_2 + 1 \leq 2(\|\mathbf{x}_k\|_2 + 1)$), we obtain

$$(T2) \leq \sqrt{\lambda_{\max}(\mathbf{A})} L\alpha \mathbb{E}_k [\|\mathbf{x}_{k-t} - \mathbf{x}^*\|_{\mathbf{A}}^2] + 4\sqrt{\lambda_{\max}(\mathbf{A})} L\alpha (\|\mathbf{x}_k\|_2 + 1)^2. \quad (\text{B.17})$$

Since $t \geq 1$, we may upper bound α by αt in the final aggregation.

Bound for (T3)+(T4). First note that Assumption 2 implies, for any $\mathbf{y}$,

$$\|\mathbf{F}(\mathbf{x}_k, \mathbf{y}) - \mathbf{F}(\mathbf{x}_{k-t}, \mathbf{y})\|_2 \leq L_1 \|\mathbf{x}_k - \mathbf{x}_{k-t}\|_2 \leq L \|\mathbf{x}_k - \mathbf{x}_{k-t}\|_2,$$

and similarly

$$\|\bar{\mathbf{F}}(\mathbf{x}_k) - \bar{\mathbf{F}}(\mathbf{x}_{k-t})\|_2 \leq L \|\mathbf{x}_k - \mathbf{x}_{k-t}\|_2.$$

Thus

$$\|\mathbf{F}(\mathbf{x}_k, \mathbf{y}_k) - \mathbf{F}(\mathbf{x}_{k-t}, \mathbf{y}_k)\|_{\mathbf{A}} + \|\bar{\mathbf{F}}(\mathbf{x}_k) - \bar{\mathbf{F}}(\mathbf{x}_{k-t})\|_{\mathbf{A}} \leq 2\sqrt{\lambda_{\max}(\mathbf{A})} L \|\mathbf{x}_k - \mathbf{x}_{k-t}\|_2.$$

Then by Cauchy–Schwarz and Lemma 4 (using (B.9)),

$$\begin{aligned} (T3) + (T4) &\leq \mathbb{E}_k [\|\mathbf{x}_{k-t} - \mathbf{x}^*\|_{\mathbf{A}}] \cdot 2\sqrt{\lambda_{\max}(\mathbf{A})} L \|\mathbf{x}_k - \mathbf{x}_{k-t}\|_2 \\ &\leq \mathbb{E}_k [\|\mathbf{x}_{k-t} - \mathbf{x}^*\|_{\mathbf{A}}] \cdot 4\sqrt{\lambda_{\max}(\mathbf{A})} L^2 \alpha t (\|\mathbf{x}_{k-t}\|_2 + 1). \end{aligned}$$

Using $\|\mathbf{x}_{k-t}\|_2 + 1 \leq 2(\|\mathbf{x}_k\|_2 + 1)$ and then $ab \leq \frac{1}{2}(a^2 + b^2)$, we get

$$(T3) + (T4) \leq 2\lambda_{\max}(\mathbf{A}) L^2 \alpha t \mathbb{E}_k [\|\mathbf{x}_{k-t} - \mathbf{x}^*\|_{\mathbf{A}}^2] + 8\lambda_{\max}(\mathbf{A}) L^2 \alpha t (\|\mathbf{x}_k\|_2 + 1)^2. \quad (\text{B.18})$$

Then we have by Lemma 2,

$$\|\mathbf{x}_k\|_2 \leq \|\mathbf{x}_k - \mathbf{x}^*\|_2 + \|\mathbf{x}^*\|_2 \leq \frac{1}{\sqrt{\lambda_{\min}^+(\mathbf{A})}} \|\mathbf{x}_k - \mathbf{x}^*\|_{\mathbf{A}} + \|\Pi_{\mathbf{A}}^\perp(\mathbf{x}_0 - \mathbf{x}^*)\|_2 + \|\mathbf{x}^*\|_2.$$

Thus, by $(a+b)^2 \leq 2a^2 + 2b^2$,

$$(\|\mathbf{x}_k\|_2 + 1)^2 \leq \frac{2}{\lambda_{\min}^+(\mathbf{A})} \|\mathbf{x}_k - \mathbf{x}^*\|_{\mathbf{A}}^2 + 4 \left(\|\Pi_{\mathbf{A}}^\perp(\mathbf{x}_0 - \mathbf{x}^*)\|_2^2 + (\|\mathbf{x}^*\|_2 + 1)^2 \right) = \frac{2}{\lambda_{\min}^+(\mathbf{A})} \|\mathbf{x}_k - \mathbf{x}^*\|_{\mathbf{A}}^2 + 4\mathcal{C}_*^2. \quad (\text{B.19})$$

Putting All Together. Plugging (B.19) into (B.16) and (B.18), and also upper bounding the α factors in (B.17) by αt (since $t \geq 1$), we obtain that (T1)–(T4) are each bounded by a constant multiple of $\frac{\lambda_{\max}(\mathbf{A})L^2}{\lambda_{\min}^+(\mathbf{A})} \alpha t (\mathbb{E}_k \|\mathbf{x}_k - \mathbf{x}^*\|_{\mathbf{A}}^2 + \lambda_{\max}(\mathbf{A})\mathcal{C}_*^2)$. This then gives (B.11). $\square$

Step 3: bounding term (c).

Lemma 6 (Squared increment bound). *For any $k \geq 0$,*

$$\mathbb{E} [\|\mathbf{F}(\mathbf{x}_k, \mathbf{y}_k)\|_{\mathbf{A}}^2] \leq \frac{4\lambda_{\max}(\mathbf{A})L^2}{\lambda_{\min}^+(\mathbf{A})} \mathbb{E} [\|\mathbf{x}_k - \mathbf{x}^*\|_{\mathbf{A}}^2] + 8\lambda_{\max}(\mathbf{A})L^2\mathcal{C}_*^2.$$

Proof. By Lemma 1,

$$\|\mathbf{F}(\mathbf{x}_k, \mathbf{y}_k)\|_{\mathbf{A}}^2 \leq \lambda_{\max}(\mathbf{A}) L^2 (\|\mathbf{x}_k\|_2 + 1)^2.$$

Use (B.19) to bound $(\|\mathbf{x}_k\|_2 + 1)^2$ by $\frac{2}{\lambda_{\min}^+(\mathbf{A})} \|\mathbf{x}_k - \mathbf{x}^*\|_{\mathbf{A}}^2 + 4\mathcal{C}_*^2$, then take expectations and simplify constants. $\square$

Step 4: Putting All Together. Substitute (B.7), Lemma 5 and Lemma 6 into terms (a)–(c). For $k \geq t$,

$$\begin{aligned} \mathbb{E}_k[\|\mathbf{x}_{k+1} - \mathbf{x}^*\|_{\mathbf{A}}^2] &\leq (1 - 2\kappa\alpha) \mathbb{E}_k[\|\mathbf{x}_k - \mathbf{x}^*\|_{\mathbf{A}}^2] + 2\alpha \cdot \frac{64 \lambda_{\max}(\mathbf{A}) L^2}{\lambda_{\min}^+(\mathbf{A})} \alpha t \left(\mathbb{E}_k[\|\mathbf{x}_k - \mathbf{x}^*\|_{\mathbf{A}}^2] + \lambda_{\max}(\mathbf{A}) \mathcal{C}_\star^2 \right) \\ &\quad + \alpha^2 \left(\frac{4 \lambda_{\max}(\mathbf{A}) L^2}{\lambda_{\min}^+(\mathbf{A})} \mathbb{E}_k[\|\mathbf{x}_k - \mathbf{x}^*\|_{\mathbf{A}}^2] + 8 \lambda_{\max}(\mathbf{A}) L^2 \mathcal{C}_\star^2 \right). \end{aligned}$$

Since $t \geq 1$, we have $\alpha^2 \leq \alpha^2 t$ and thus we may upper bound the last line by an $\alpha^2 t$ -term. Collecting the coefficients (and noting $2 \cdot 64 = 128$) yields

$$\begin{aligned} \mathbb{E}_k[\|\mathbf{x}_{k+1} - \mathbf{x}^*\|_{\mathbf{A}}^2] &\leq \left(1 - 2\kappa\alpha + \frac{130 \lambda_{\max}(\mathbf{A}) L^2}{\lambda_{\min}^+(\mathbf{A})} \alpha^2 t \right) \mathbb{E}_k[\|\mathbf{x}_k - \mathbf{x}^*\|_{\mathbf{A}}^2] \\ &\quad + \frac{130 \lambda_{\max}(\mathbf{A}) L^2}{\lambda_{\min}^+(\mathbf{A})} \alpha^2 t \cdot \lambda_{\max}(\mathbf{A}) \mathcal{C}_\star^2. \end{aligned} \tag{B.20}$$

Under the stepsize choice (B.4), we have

$$\frac{130 \lambda_{\max}(\mathbf{A}) L^2}{\lambda_{\min}^+(\mathbf{A})} \alpha t \leq \kappa,$$

hence the coefficient in (B.20) satisfies

$$1 - 2\kappa\alpha + \frac{130 \lambda_{\max}(\mathbf{A}) L^2}{\lambda_{\min}^+(\mathbf{A})} \alpha^2 t \leq 1 - \kappa\alpha.$$

Therefore,

$$\mathbb{E}_k[\|\mathbf{x}_{k+1} - \mathbf{x}^*\|_{\mathbf{A}}^2] \leq (1 - \kappa\alpha) \mathbb{E}_k[\|\mathbf{x}_k - \mathbf{x}^*\|_{\mathbf{A}}^2] + \frac{260 \lambda_{\max}(\mathbf{A}) L^2}{\lambda_{\min}^+(\mathbf{A})} \alpha^2 t \cdot \mathcal{C}_\star^2,$$

where we used $\lambda_{\max}(\mathbf{A}) \mathcal{C}_\star^2 \leq \frac{2 \lambda_{\max}(\mathbf{A})}{\lambda_{\min}^+(\mathbf{A})} \mathcal{C}_\star^2$ up to constants and absorbed into 260. Finally, take total expectation of both sides to obtain (B.5). $\square$

B.3 Proof of Theorem 3.1

Now we are ready to prove the evaluation result with behaviour policy μ and target policy π .

Throughout this section, we simply write ϕ^π by ϕ and index the state and action spaces as $\mathcal{S} = \{s_1, \dots, s_n\}$ and $\mathcal{A} = \{a_1, \dots, a_m\}$. Let $d_\mu \in \mathbb{R}^{mn}$ denote the stationary distribution of (S, A) under μ (allowing zero components), and define

$$\begin{aligned} \mathbf{D} &:= \text{diag}(\bar{d}_\mu(s_1, a_1), \dots, \bar{d}_\mu(s_n, a_m)) \in \mathbb{R}^{mn \times mn}, \\ \mathbf{\Phi} &:= [\phi(s_1, a_1), \dots, \phi(s_n, a_m)]^\top \in \mathbb{R}^{mn \times d}. \end{aligned}$$

Note that $\mathbf{D}$ (and hence $\mathbf{\Phi}^\top \mathbf{D} \mathbf{\Phi}$) is allowed to be singular. We also use $\mathcal{H} := \mathcal{H}^\pi$ to denote the Bellman evaluation operator:

$$[\mathcal{H}(Q)](s, a) := r(s, a) + \gamma \sum_{s' \in \mathcal{S}} P(s'|s, a) \sum_{a' \in \mathcal{A}} \pi(a'|s') Q(s', a').$$

Equivalently, define the state-action transition kernel $(\mathbf{P}_\pi Q)(s, a) := \sum_{s'} P(s'|s, a) \sum_{a'} \pi(a'|s') Q(s', a')$ so that

$$\mathcal{H}(Q) = \mathbf{r} + \gamma \mathbf{P}_\pi Q, \quad .$$

Our first step is a convergence analysis of the inner-loop TD update:

Step 1: Local Convergence Inner-loop TD with fixed target. Fix a target network parameter $w \in \mathbb{R}^d$ and consider the TD-type update driven by a Markov trajectory (S_k, A_k, S_{k+1}) under μ :

$$\xi_{k+1} = \xi_k + \alpha \phi(S_k, A_k) \left(r(S_k, A_k) + \gamma \sum_{a' \in \mathcal{A}} \pi(a'|S_{k+1}) [\phi(S_{k+1}, a')^\top w] - \phi(S_k, A_k)^\top \xi_k \right).$$

This is a Markovian SA instance of (B.1) with $\mathbf{y}_k = (S_k, A_k, S_{k+1})$ and

$$\begin{aligned} \mathbf{F}(\xi, (s, a, s')) &:= \phi(s, a) \left(-\phi(s, a)^\top \xi + r(s, a) + \gamma \sum_{a' \in \mathcal{A}} \pi(a'|s') [\phi(s', a')^\top w] \right), \\ \bar{\mathbf{F}}(\xi) &:= \mathbb{E}_{\substack{(S, A) \sim d_\mu \\ S' \sim P(\cdot|S, A)}} \left[\mathbf{F}(\xi, (S, A, S')) \right] = \Phi^\top \mathbf{D} \left(-\Phi \xi + \mathcal{H}(\lceil \Phi w \rceil) \right). \end{aligned}$$

Now it remains to verify Theorem B.1.

Verification of Theorem B.1. With $\mathbf{A} := \Phi^\top \mathbf{D} \Phi \succeq \mathbf{0}$, we have:

Condition 1. By the aperiodic assumption, we have the Markov chain $\{(S_k, A_k, S_{k+1})\}$ induced by μ is geometrically mixing on its support:

$$\max_{y \in \mathcal{Y}} \|P^k(y, \cdot) - \nu(\cdot)\|_{\text{TV}} \leq C\rho^k,$$

where ν is the stationary distribution over $\mathcal{Y} := \{(s, a, s') : d_\mu(s, a)P(s'|s, a) > 0\}$. Consequently, the mixing time in Theorem B.1 satisfies the explicit bound

$$\tau_\alpha := \min \left\{ k \geq 0 : \max_{y \in \mathcal{Y}} \|P^k(y, \cdot) - \nu(\cdot)\|_{\text{TV}} \leq \alpha \right\} \leq \left\lceil \frac{\log(C/\alpha)}{\log(1/\rho)} \right\rceil. \quad (\text{B.21})$$

Condition 2. By our Assumption, we have $\|\phi(s, a)\|_2 \leq 1$ for all (s, a) . Then for all (s, a, s') ,

$$\|\mathbf{F}(\xi_1, (s, a, s')) - \mathbf{F}(\xi_2, (s, a, s'))\|_2 = \|\phi(s, a)\phi(s, a)^\top (\xi_2 - \xi_1)\|_2 \leq \|\phi(s, a)\|_2^2 \|\xi_1 - \xi_2\|_2 \leq \|\xi_1 - \xi_2\|_2.$$

Moreover, using $r \in [0, 1]$ and $|\lceil \phi^\top w \rceil| \leq 1/(1 - \gamma)$,

$$\|\mathbf{F}(\mathbf{0}, (s, a, s'))\|_2 \leq \|\phi(s, a)\|_2 \left(1 + \gamma B \right) \leq 1 + 1/(1 - \gamma).$$

Thus we may take $L_1 = 1$ and $L_2 = 2/(1 - \gamma)$.

Condition 3. Let $\xi^*(w)$ be any solution to $\bar{\mathbf{F}}(\xi) = \mathbf{0}$:

$$\Phi^\top \mathbf{D} \Phi \xi^*(w) = \Phi^\top \mathbf{D} \mathcal{H}(\lceil \Phi w \rceil).$$

Then for any $\xi \in \mathbb{R}^d$,

$$\begin{aligned} \langle \xi - \xi^*, \bar{\mathbf{F}}(\xi) \rangle_{\mathbf{A}} &= \langle \xi - \xi^*, \bar{\mathbf{F}}(\xi) - \bar{\mathbf{F}}(\xi^*) \rangle_{\mathbf{A}} = \langle \xi - \xi^*, \mathbf{A}(\xi^* - \xi) \rangle_{\mathbf{A}} \\ &= -(\xi - \xi^*)^\top \mathbf{A}^2 (\xi - \xi^*) \leq -\lambda_{\min}^+(\mathbf{A}) (\xi - \xi^*)^\top \mathbf{A} (\xi - \xi^*) = -\lambda_{\min}^+(\mathbf{A}) \|\xi - \xi^*\|_{\mathbf{A}}^2. \end{aligned}$$

Hence Condition 3 holds with $\kappa = \lambda_{\min}^+(\mathbf{A})$.

Condition 4. Noticing that for any $(s, a, s') \in \mathcal{Y}$, we have $\mathbf{F}(\xi, (s, a, s')) \propto \phi(s, a)$ and moreover, since (s, a) belongs to the support of $\bar{d}_\mu$ on $\mathcal{Y}$, we have $d_\mu(s, a) > 0$. Noticing that $\text{Range}(\mathbf{A}) = \text{span}\{\phi(s, a) : d_\mu(s, a) > 0\}$, we have $\phi(s, a) \in \text{Range}(\mathbf{A})$ for all (s, a) appearing in $\mathcal{Y}$. Therefore $\mathbf{F}(\xi, (s, a, s')) \in \text{Range}(\mathbf{A})$ for all $(s, a, s') \in \mathcal{Y}$. $\square$

Now we can apply Theorem B.1 for all $k \geq \tau_\alpha$ to arrive at for $c_0 = \lambda_{\min}^+(\mathbf{A})$, $c_1 \asymp \frac{1}{(1-\gamma)^2 \lambda_{\min}^+(\mathbf{A})}$

$$\mathbb{E}[\|\xi_{k+1} - \xi^*(w)\|_{\mathbf{A}}^2] \leq (1 - c_0\alpha) \mathbb{E}[\|\xi_k - \xi^*(w)\|_{\mathbf{A}}^2] + c_1\alpha^2\tau_\alpha.$$

By recursion, for any $K \geq \tau_\alpha$, we have then

$$\begin{aligned} \mathbb{E}[\|\xi_K - \xi^*(w)\|_{\mathbf{A}}^2] &\leq (1 - c_0\alpha)^K \mathbb{E}[\|\xi_0 - \xi^*(w)\|_{\mathbf{A}}^2] + \underbrace{c_1/c_0}_{\asymp ((1-\gamma)\lambda_{\min}^+(\mathbf{A}))^{-2}} \alpha\tau_\alpha. \end{aligned}$$

Remark 4 (An explicit choice of (α, K) for inner-loop accuracy ϵ). Fix any $\epsilon \in (0, 1)$ and define $V_0(w) := \mathbb{E}[\|\xi_0 - \xi^*(w)\|_{\mathbf{A}}^2]$. Using $(1 - c_0\alpha)^K \leq \exp(-c_0\alpha K)$, it suffices to ensure

$$\exp(-c_0\alpha K) V_0(w) \leq \frac{\epsilon}{2}, \quad \frac{c_1}{c_0} \alpha \tau_\alpha \leq \frac{\epsilon}{2}.$$

Under the geometric mixing bound (B.21), a convenient explicit choice is

$$\alpha := \min \left\{ \frac{\lambda_{\min}^+(\mathbf{A})}{130 \max(L_1, L_2)^2} \cdot \frac{\log(1/\rho)}{2 \log(2C/\epsilon)}, \quad \frac{c_0}{4c_1} \epsilon \cdot \frac{\log(1/\rho)}{\log(2C/\epsilon)} \right\}, \quad (\text{B.22})$$

and

$$K := \left\lceil \frac{1}{c_0\alpha} \log \left(\frac{2V_0(w)}{\epsilon} \right) \right\rceil + \tau_\alpha. \quad (\text{B.23})$$

With (B.22)–(B.23), we have $\mathbb{E}[\|\xi_K - \xi^*(w)\|_{\mathbf{A}}^2] \leq \epsilon^2$. In particular, since $\tau_\alpha = O(\log(1/\alpha))$ by (B.21), this gives an inner-loop complexity of order

$$K = O\left(\frac{\log(1/\epsilon) \log(1/\alpha)}{\epsilon^2 \lambda_{\min}^+(\mathbf{A})} \right)$$

up to problem-dependent constants.

Convergence of the target variables. Now we are ready to complete the proof of Theorem 3.1 based on the above inner loop bound and contraction of Bellman operator. The following proof is inspired by the arguments of Chen et al. (2023) under ℓ_∞ norm, but here we make a careful modification change of it to the on-policy L^2 norm, which is the key step to tackle the partial coverage of μ and utilize previous convergence result under $\mathbf{A}$.

For clarity, we denote $\{w_k\}_{k \geq 0}$ be the target-network sequence generated by the outer loop update, and at outer iteration $k \geq 1$ define the returned estimate

$$\hat{Q}_k := \Phi \xi_K(w_{k-1}) \in \mathbb{R}^{mn}.$$

Let $Q^* := Q^\pi$ denote the true evaluation Q -function under π . We denote the joint (s, a) stationary distribution of π by $\bar{d}_\pi$, which satisfies

$$\bar{d}_\pi^\top \mathbf{P}_\pi = \bar{d}_\pi^\top.$$

Using $\mathcal{H}(Q) = \mathbf{r} + \gamma \mathbf{P}_\pi Q$ and the entrywise inequality $|\mathbf{P}_\pi \mathbf{v}| \leq \mathbf{P}_\pi |\mathbf{v}|$, we have

$$|\mathcal{H}(\lceil \hat{Q}_{k-1} \rceil) - \mathcal{H}(Q^*)| = \gamma |\mathbf{P}_\pi (\lceil \hat{Q}_{k-1} \rceil - Q^*)| \leq \gamma \mathbf{P}_\pi |\hat{Q}_{k-1} - Q^*|.$$

Therefore, for each $k \geq 1$, we have pointwisely,

$$\begin{aligned} |\hat{Q}_k - Q^*| &= \left| \mathcal{H}(\lceil \hat{Q}_{k-1} \rceil) - \mathcal{H}(Q^*) + \hat{Q}_k - \Phi \xi^*(w_{k-1}) + \mathcal{H}(\lceil \hat{Q}_{k-1} \rceil) - \Phi \xi^*(w_{k-1}) \right| \\ &\leq \gamma \mathbf{P}_\pi |\hat{Q}_{k-1} - Q^*| + |\hat{Q}_k - \Phi \xi^*(w_{k-1})| + \underbrace{|\mathcal{H}(\lceil \hat{Q}_{k-1} \rceil) - \Phi \xi^*(w_{k-1})|}_{\leq \mathcal{E}_{\text{approx}} + |\Phi(\xi' - \xi^*(w_{k-1}))|}. \end{aligned}$$

Where ξ' is the best-approximation representation in Assumption 3.

Taking squared in both sides and using weighted geometric inequality, we can have pointwisely

$$\begin{aligned} |\hat{Q}_k - Q^*|^2 &\leq \gamma \underbrace{(P_\pi |\hat{Q}_{k-1} - Q^*|)^2}_{\leq P_\pi |\hat{Q}_{k-1} - Q^*|^2 \text{ by Cauchy-Schwartz}} + \frac{16}{(1-\gamma)^2} |\hat{Q}_k - \Phi \xi^*(w_{k-1})|^2 \\ &\quad + 4\mathcal{E}_{\text{approx}}^2 + 4|\Phi(\xi' - \xi^*(w_{k-1}))|^2. \end{aligned}$$

Left-multiplying by $\bar{d}_\pi^\top$ and using $\bar{d}_\pi^\top P_\pi = \bar{d}_\pi^\top$ then yields the L^2 recursion

$$\begin{aligned} \mathbb{E}_{(s,a) \sim d_\pi} [|\hat{Q}_k(s,a) - Q^*(s,a)|^2] &\leq \gamma \mathbb{E}_{\bar{d}_\pi} [|\hat{Q}_{k-1}(s,a) - Q^*(s,a)|^2] + \frac{16}{(1-\gamma)^2} \bar{d}_\pi^\top [|\hat{Q}_k - \Phi \xi^*(w_{k-1})|^2] \\ &\quad + 4\mathcal{E}_{\text{approx}}^2 + 4\bar{d}_\pi^\top |\Phi(\xi' - \xi^*(w_{k-1}))|^2. \end{aligned}$$

Bounding the Second Term. For the second term, we have by

$$|\phi(s,a)^\top (\xi_K(w_{k-1}) - \xi^*(w_{k-1}))| \leq \|\phi(s,a)\|_{\mathbf{A}^{-1}} \|\xi_K(w_{k-1}) - \xi^*(w_{k-1})\|_{\mathbf{A}},$$

for any $\mathbf{A}$, with

$$\|\phi(s,a)\|_{\mathbf{A}^{-1}} = \lim_{\lambda \rightarrow 0^+} \|\phi(s,a)\|_{(\mathbf{A} + \lambda \mathbf{I})^{-1}} = \begin{cases} \|\phi(s,a)\|_{\mathbf{A}^\dagger}, & \text{if } \phi(s,a) \in \text{Range}(\mathbf{A}), \\ +\infty, & \text{otherwise.} \end{cases}$$

Hence,

$$\mathbb{E}_{\bar{d}_\pi} |\Phi(\xi_K(w_{k-1}) - \xi^*(w_{k-1}))|^2 \leq \mathbb{E}_{\bar{d}_\pi} [\|\phi(s,a)\|_{\mathbf{A}^{-1}}^2] \cdot \|\xi_K(w_{k-1}) - \xi^*(w_{k-1})\|_{\mathbf{A}}^2$$

Thus, if the inner-loop is run with accuracy $\mathbb{E}[\|\xi_K(w) - \xi^*(w)\|_{\mathbf{A}}^2] \leq \epsilon^2$ (i.e. with the parameter setting as we discussed in Remark 4), then

$$\mathbb{E} \left[\bar{d}_\pi^\top |\Phi(\xi_K(w_{k-1}) - \xi^*(w_{k-1}))|^2 \right] \leq \epsilon^2 \cdot \mathbb{E}_{(s,a) \sim \bar{d}_\pi} [\|\phi(s,a)\|_{\mathbf{A}^{-1}}^2].$$

Bounding the Last Term. Similarly, by definition of $\xi^*(w_{k-1})$, we have

$$\begin{aligned} \Phi^\top D \Phi \xi^*(w_{k-1}) &= \Phi^\top D \mathcal{H}(\lceil \hat{Q}_{k-1} \rceil) = \Phi^\top D \Phi \xi' + O(\mathcal{E}_{\text{approx}}) \\ \implies \|\xi^*(w_{k-1}) - \xi'\|_{\mathbf{A}}^2 &= (\xi^*(w_{k-1}) - \xi')^\top \mathbf{A} \mathbf{A}^{-1} \mathbf{A} (\xi^*(w_{k-1}) - \xi') = O(\mathcal{E}_{\text{approx}}^2 / \lambda_{\min}^+(\mathbf{A})) \end{aligned}$$

thus

$$\bar{d}_\pi^\top |\Phi(\xi' - \xi^*(w_{k-1}))|^2 \leq \mathbb{E}_{\bar{d}_\pi} [\|\phi(s,a)\|_{\mathbf{A}^{-1}}^2] \cdot \underbrace{\|\xi' - \xi^*(w_{k-1})\|_{\mathbf{A}}^2}_{=O(\mathcal{E}_{\text{approx}}^2)}$$

Putting All Together. Combining all above bounds together and take recursion, we have any $T \geq 1$,

$$\begin{aligned} \mathbb{E}_{(s,a) \sim d_\pi} [|\hat{Q}_T(s,a) - Q^*(s,a)|^2] &\leq \gamma^T \mathbb{E}_{(s,a) \sim d_\pi} [|\hat{Q}_0(s,a) - Q^*(s,a)|^2] + \\ &\quad \frac{\mathbb{E}_{(s,a) \sim d_\pi} [\|\phi(s,a)\|_{\mathbf{A}^{-1}}^2]}{1-\gamma} (\epsilon^2 + \mathcal{E}_{\text{approx}}^2) + \mathcal{E}_{\text{approx}}^2, \end{aligned}$$

as desired.

B.4 Proof of Corollary 1

By our Assumption 2, we have

$$\begin{aligned} \mathbb{E}_{(s,a) \sim \bar{d}_\pi} |\hat{Q}_T(s,a) - Q^\pi(s,a)|^2 &= \sum_{s,a} \bar{d}_\pi(s,a) |\langle \phi^\pi(s,a), \hat{\xi}_T - \xi^* \rangle|^2 \\ &= (\hat{\xi}_T - \xi^*)^\top \underbrace{\left[\sum_{s,a} \bar{d}_\pi(s,a) \phi^\pi(s,a) \phi^\pi(s,a)^\top \right]}_{=\mathbf{A}_\pi} (\hat{\xi}_T - \xi^*) = \|\hat{\xi}_T - \xi^*\|_{\mathbf{A}_\pi}^2. \end{aligned}$$

Then taking expectation in both sides and applying Theorem 1 leads to the desired result.

B.5 Proof of Theorem 3.2

First, by our design of Algorithm 3 and (3.3), we have for every k, t , denote $\mathcal{F}_t^{(k)}$ the historical information up to the time we are computing the gradient of the k -th episode, t -th policy $\pi_t^{(k)}$, then it holds that

$$\mathbb{E}[\|\nabla J_\rho(\pi_t^{(k)}) - \hat{\nabla} J_\rho(\pi_t^{(k)})\|_2^2 | \mathcal{F}_t^{(k)}] \lesssim \bar{\epsilon}^2. \quad (\text{B.24})$$

Now if we use the Θ -parametrization of π and re-index the parameter corresponding to the policy at the k -th episode, t -th update as θ_m , then with the notation $f(\theta) := J_\rho(\pi_\theta)$, we have by Assumption 4

$$\begin{aligned} f(\theta_{m+1}) &\leq f(\theta_m) + \langle \nabla f(\theta_m), \theta_{m+1} - \theta_m \rangle + \frac{L}{2} \|\theta_{m+1} - \theta_m\|_2^2 \\ &= f(\theta_m) - \eta \langle \nabla f(\theta_m), \hat{\nabla} f(\theta_m) \rangle + \frac{L\eta^2}{2} \|\hat{\nabla} f(\theta_m)\|_2^2 \end{aligned}$$

Now taking conditional expectation over both sides, we have by (B.24),

$$\begin{aligned} \mathbb{E}[f(\theta_{m+1}) | \mathcal{F}_m] &\leq f(\theta_m) - \eta \|\nabla f(\theta_m)\|_2^2 + \frac{L\eta^2}{2} \mathbb{E}[\|\hat{\nabla} f(\theta_m)\|_2^2 | \mathcal{F}_m] + \eta \mathbb{E}[\langle \nabla f(\theta_m), \nabla f(\theta_m) - \hat{\nabla} f(\theta_m) \rangle | \mathcal{F}_m] \\ &\leq f(\theta_m) - \frac{\eta}{2} \|\nabla f(\theta_m)\|_2^2 + \frac{L\eta^2}{2} \mathbb{E}[\|\hat{\nabla} f(\theta_m) \pm \nabla f(\theta_m)\|_2^2 | \mathcal{F}_m] + 4\eta \mathbb{E}[\|\nabla f(\theta_m) - \hat{\nabla} f(\theta_m)\|_2^2 | \mathcal{F}_m] \\ &\leq f(\theta_m) - \frac{\eta}{2} \|\nabla f(\theta_m)\|_2^2 + 16\eta\bar{\epsilon}^2 + 4L\eta^2\bar{\epsilon}^2 + 2L\eta^2 \|\nabla f(\theta_m)\|_2^2 \\ &\leq f(\theta_m) + \underbrace{\left(2L\eta^2 - \frac{\eta}{2}\right)}_{\leq -\frac{\eta}{4} \text{ by } \eta \leq \frac{1}{8L}} \|\nabla f(\theta_m)\|_2^2 + 16\eta\bar{\epsilon}^2 + 4L\eta^2\bar{\epsilon}^2. \end{aligned}$$

This gives when $\eta \leq 1/8L$,

$$\frac{\eta}{4} \mathbb{E}[\|\nabla f(\theta_m)\|_2^2] \leq \mathbb{E}[f(\theta_m) - f(\theta_{m+1})] + O(\eta\bar{\epsilon}^2).$$

Now taking summation from $m = 1$ to M and dividing both sides by ηM , we have

$$\frac{1}{M} \sum_{m=1}^M \mathbb{E}[\|\nabla f(\theta_m)\|_2^2] \lesssim \frac{\mathbb{E}[f(\theta_1) - f(\theta^*)]}{M\eta} + \bar{\epsilon}^2,$$

as desired.

C Optimal Buffer Selection under Partial Coverage

The convergence analysis in §3.2 characterizes the quality of the functional critic through the evaluation trust metric $\mathcal{C}(\pi, \mu)$ (Definition 1), which is defined for a *population* behavior measure μ . In many practical deployments, however, the learner does not directly choose a behavior policy: it is given a fixed offline buffer $\mathcal{B}$ of transitions, possibly produced by a mixture of historical policies, and must decide how to use it. This section asks two natural questions of the trust-coverage framework:

1. Among all sub-samples $S \subseteq \mathcal{B}$, which one yields the smallest evaluation error of the functional critic learned by Algorithm 2?
2. Does the answer collapse to the trivial $S = \mathcal{B}$, or is there a non-trivial trade-off that makes *dropping* certain transitions strictly beneficial?

To keep the discussion focused on the selection mechanism itself, we specialize to a *fixed* target policy π ; the extension to evolving target policies, where selection couples with the dual-layer mechanism of Algorithm 3, is discussed at the end.

The answer to (1) and (2) will turn out to come from a **two-layer error decomposition** of the algorithm output: an LSTD-style approximation error that decreases as $|S|$ grows, and a TD-to-LSTD optimization gap that grows with $|S|$ for a fixed sample budget. The optimal S^* balances these two layers.

C.1 Setup: Subset Selection on a Fixed Buffer

Fix the target policy π and let $\mathcal{B} = \{(s_i, a_i, r_i, s'_i)\}_{i=1}^N$ be a fixed offline buffer. Write $\phi_i := \phi^\pi(s_i, a_i) \in \mathbb{B}^d$ for the functional feature at each transition. A *selection* on $\mathcal{B}$ is a subset

$$S \subseteq [N], \quad |S| = m,$$

and the associated empirical measure μ_S is the uniform distribution on S . Replacing the behavior policy μ in Algorithm 2 by μ_S amounts to drawing each inner-loop transition uniformly at random (with replacement) from S . Because μ_S is a fixed discrete distribution, the inner-loop transitions are i.i.d. samples from μ_S , the Markov mixing condition (Assumption 1) is trivially satisfied with mixing time one, and the per-step Markovian-noise hypothesis of Theorem 3.1 applies verbatim with $\mu \leftarrow \mu_S$.

Empirical feature covariance. The discrete object that drives the entire analysis is the *unnormalized* feature covariance of the selected subset:

$$A_S := \sum_{i \in S} \phi_i \phi_i^\top \in \mathbb{R}^{d \times d}.$$

We deliberately do *not* normalize by $|S|$: a larger S produces a larger A_S (in the PSD order), so A_S^{-1} shrinks, and the trust metric below inherits an explicit $|S|$ dependence that captures the sample-count contribution to coverage. The Moore–Penrose pseudoinverse convention of Definition 1 applies when A_S is singular.

C.2 The Trust Metric on a Subset

Specializing Definition 1 to the empirical measure μ_S , and using A_S in place of the population feature covariance, the subset trust metric is

$$\bar{\mathcal{C}}(\pi, S) := \mathbb{E}_{\bar{d}_\pi} [\|\phi^\pi(s, a)\|_{A_S^{-1}}^2] = \text{tr}(\mathbf{A}_\pi A_S^{-1}), \quad (\text{C.1})$$

where $\mathbf{A}_\pi := \mathbb{E}_{\bar{d}_\pi} [\phi^\pi \phi^{\pi^\top}]$ is the on-policy feature covariance from §3.2. The metric is in the *same form* as Definition 1 (inverse-norm expectation, squared), but A_S being unnormalized means $\bar{\mathcal{C}}(\pi, S)$ inherits a $1/|S|$ rate factor from the buffer.

Relationship to Definition 1. If we were to apply Definition 1 directly with $\mu \leftarrow \mu_S$, the empirical covariance would be the *normalized* $\frac{1}{|S|} A_S$, giving $\mathcal{C}(\pi, \mu_S) = |S| \cdot \bar{\mathcal{C}}(\pi, S)$. The unnormalized form (C.1) is the one that arises naturally in the offline-LSTD error bound below; the normalization is absorbed into the sample-count factor of the TD-to-LSTD gap.

An intuition trap. A naïve reading of (C.1) suggests selecting samples i with small $\|\phi_i\|$ to make the right-hand side small. This is precisely backwards. Samples whose features ϕ_i align with the directions of high $\mathbf{A}_\pi$ -mass are the ones that, when added to S , most strongly expand A_S along the directions relevant to $\bar{d}_\pi$, thereby *reducing* the inverse-norm expectation. The right intuition is *include samples with high $\mathbf{A}_\pi$ -aligned leverage; drop samples whose feature is nearly orthogonal to $\mathbf{A}_\pi$'s spectrum*.

C.3 Two-Layer Error Decomposition

The output $\hat{\xi}_S^{\text{TD}}$ of Algorithm 2 run on μ_S for K inner iterations and T outer iterations differs from the true target ξ^* along two complementary error channels: the LSTD-style approximation error of the empirical fixed point ξ_S^* , and the TD optimization gap between $\hat{\xi}_S^{\text{TD}}$ and ξ_S^* . We state each as a lemma and then combine.

The empirical fixed point. The inner loop of Algorithm 2, with i.i.d. transitions from μ_S and infinitely many TD steps, converges to the solution of the projected, truncated Bellman fixed-point equation

$$\xi_S^* = A_S^{-1} b_S(\xi_S^*), \quad b_S(\xi) := \sum_{i \in S} \phi_i \left[r_i + \gamma \sum_a \pi(a | s'_i) [\phi^\pi(s'_i, a)^\top \xi] \right], \quad (\text{C.2})$$

where $[\cdot]$ truncates to $[0, (1 - \gamma)^{-1}]$ as in Algorithm 2. We refer to ξ_S^* as the *LSTD fixed point on S* : it is the limit the inner loop would reach if $K \rightarrow \infty$.

Lemma 7 (LSTD Approximation Error). *Let Assumptions 2 and 3 hold for the target policy π , and suppose $A_S \succ 0$. The LSTD fixed point ξ_S^* defined in (C.2) satisfies*

$$\|\xi_S^* - \xi^*\|_{\mathbf{A}_\pi}^2 \lesssim (1 - \gamma)^{-3} \bar{\mathcal{C}}(\pi, S) + \mathcal{E}_{\text{approx}}^2.$$

[*yx:cite offline-LSTD evaluation bound; candidates: duan2020minimax, min2021variance, xiong2022nearly.*]

Lemma 8 (TD-to-LSTD Optimization Gap). *Under the same hypotheses, if Algorithm 2 is run on μ_S with K inner-loop TD steps per outer iteration and target-network iterations T taken to convergence, the iterate $\hat{\xi}_S^{\text{TD}}$ satisfies*

$$\|\hat{\xi}_S^{\text{TD}} - \xi_S^*\|_{\mathbf{A}_\pi}^2 \lesssim (1 - \gamma)^{-4} \cdot \frac{|S|}{K} \cdot \bar{\mathcal{C}}(\pi, S).$$

[*yx:verify the $(1 - \gamma)^{-4} \cdot |S|/K$ form; cite the TD-on-empirical-buffer convergence-rate reference.*]

The two lemmas combine through a weak triangle inequality ($\|x + y\|^2 \leq 2\|x\|^2 + 2\|y\|^2$) into the following theorem.

Theorem C.1 (Selection-Aware Total Error). *Let Assumptions 1–3 hold and suppose $A_S \succ 0$. Then*

$$\|\hat{\xi}_S^{\text{TD}} - \xi^*\|_{\mathbf{A}_\pi}^2 \lesssim \underbrace{(1 - \gamma)^{-3} \bar{\mathcal{C}}(\pi, S)}_{\text{LSTD bound (\S 7)}} + \underbrace{(1 - \gamma)^{-4} \frac{|S|}{K} \bar{\mathcal{C}}(\pi, S)}_{\text{TD-to-LSTD gap (\S 8)}} + \mathcal{E}_{\text{approx}}^2. \quad (\text{C.3})$$

Equivalently,

$$\|\hat{\xi}_S^{\text{TD}} - \xi^*\|_{\mathbf{A}_\pi}^2 \lesssim \left(1 + \frac{|S|/K}{1 - \gamma}\right) (1 - \gamma)^{-3} \bar{\mathcal{C}}(\pi, S) + \mathcal{E}_{\text{approx}}^2.$$

Proof. Combine Lemmas 7 and 8 via $\|\hat{\xi}_S^{\text{TD}} - \xi^*\|_{\mathbf{A}_\pi}^2 \leq 2\|\hat{\xi}_S^{\text{TD}} - \xi_S^*\|_{\mathbf{A}_\pi}^2 + 2\|\xi_S^* - \xi^*\|_{\mathbf{A}_\pi}^2$, absorb the constant 2 into the $\lesssim$ notation, and factor $\bar{\mathcal{C}}(\pi, S)$ out of the two coverage-dependent terms. $\square$

C.4 The Selection Problem and the Trade-off

Theorem C.1 prescribes a concrete selection objective. With K (the inner-loop TD budget) fixed by the algorithm configuration, the optimal subset is

$$S^* \in \arg \min_{S \subseteq [N]} \left(1 + \frac{|S|/K}{1 - \gamma}\right) \cdot \bar{\mathcal{C}}(\pi, S). \quad (\text{C.4})$$

Why selection is non-trivial. Consider the two limits of (C.4):

- **LSTD-only regime** ($K \rightarrow \infty$, the TD gap vanishes): the objective reduces to $\bar{\mathcal{C}}(\pi, S)$, which is monotone non-increasing in S under set inclusion. Hence the trivial choice $S = [N]$ (the full buffer) is optimal, and there is no selection problem.
- **Finite- K regime** (the TD gap is active): the factor $\frac{|S|/K}{1 - \gamma}$ grows linearly in $|S|$, while $\bar{\mathcal{C}}(\pi, S)$ shrinks sub-linearly when the buffer contains samples of varying quality. Specifically, dropping a sample i whose feature ϕ_i is nearly orthogonal to the $\mathbf{A}_\pi$ spectrum changes $\bar{\mathcal{C}}(\pi, S)$ by only $o(1/|S|)$ but strictly reduces the multiplicative factor by $1/(K(1 - \gamma))$. Such drops therefore strictly improve the bound.

The selection problem (C.4) is the natural mathematical expression of this trade-off: it asks the learner to retain samples that carry $\mathbf{A}_\pi$ -aligned coverage and discard those that merely inflate $|S|$ without contributing to it.

A target-aligned leverage criterion. The marginal contribution of sample i to $\bar{\mathcal{C}}(\pi, S)$ when added to a current subset S is, by the Sherman–Morrison identity,

$$\bar{\mathcal{C}}(\pi, S) - \bar{\mathcal{C}}(\pi, S \cup \{i\}) = \frac{\phi_i^\top A_S^{-1} \mathbf{A}_\pi A_S^{-1} \phi_i}{1 + \phi_i^\top A_S^{-1} \phi_i}. \quad (\text{C.5})$$

The numerator $\phi_i^\top A_S^{-1} \mathbf{A}_\pi A_S^{-1} \phi_i$ is the $\mathbf{A}_\pi$ -weighted leverage score of i relative to S — the sample’s contribution to $\bar{\mathcal{C}}(\pi, S)$ in the direction of the target distribution. Samples with small leverage (small numerator) are the natural candidates to drop in (C.4).

C.5 Empirical Fixed Point and the Asymptotic Gap to Q^π

Theorem C.1 also clarifies what the algorithm is *asymptotically converging to*. In the $K \rightarrow \infty$ limit the TD-to-LSTD gap term in (C.3) vanishes, and we recover the pure LSTD bound

$$\|\xi_S^* - \xi^*\|_{\mathbf{A}_\pi}^2 \lesssim (1 - \gamma)^{-3} \bar{\mathcal{C}}(\pi, S) + \mathcal{E}_{\text{approx}}^2. \quad (\text{C.6})$$

In the language of §C.1, ξ_S^* is the algorithm’s *approximation target*: the parameter the algorithm would deliver in the absence of TD optimization error. Equation (C.6) quantifies the price of using μ_S in place of an idealized fresh population — and it is governed by the same metric $\bar{\mathcal{C}}(\pi, S)$ that drives the finite- K bound.

Remark 5 (Two layers of gap, one metric). *Both the LSTD approximation error and the TD-to-LSTD optimization gap scale linearly in $\bar{\mathcal{C}}(\pi, S)$. Selection therefore controls both layers simultaneously: a subset S that lowers $\bar{\mathcal{C}}$ lowers both the asymptotic bias to Q^π and the finite-iteration variance, while the relative weighting between the two layers is set by $|S|/K$.*

C.6 Solving the Selection Problem

Solving (C.4) exactly is combinatorial: even the cardinality-restricted inner problem $\min_{|S|=m} \bar{\mathcal{C}}(\pi, S)$ is NP-hard (Das and Kempe, 2008). The natural decomposition splits along the two factors in the objective:

$$\min_{S \subseteq [N]} \left(1 + \frac{|S|/K}{1-\gamma}\right) \bar{\mathcal{C}}(\pi, S) = \min_m \left(1 + \frac{m/K}{1-\gamma}\right) \underbrace{\min_{|S|=m} \bar{\mathcal{C}}(\pi, S)}_{\text{cardinality-constrained}}. \quad (\text{C.7})$$

The outer minimization over m is a 1-D scan. The inner minimization admits a $(1 - 1/e)$ -approximation via greedy submodular maximization, which we develop next. Two auxiliary routes — a continuous convex relaxation and a spectral-sparsification structural result — are discussed afterward.

Greedy submodular solver (main route). The starting point is a classical submodularity property of the $\mathbf{A}_\pi$ -weighted prediction variance.

Lemma 9 (Submodularity of $-\bar{\mathcal{C}}$). *Let $\mathbf{A}_\pi \succeq 0$ and let $S_0 \subseteq [N]$ be any anchor with $A_{S_0} \succ 0$. The set function $g(S) := \bar{\mathcal{C}}(\pi, S_0) - \bar{\mathcal{C}}(\pi, S)$, defined on subsets $S \supseteq S_0$, is non-negative, monotone non-decreasing, and submodular in S .*

Proof. Diagonalize $\mathbf{A}_\pi = \sum_k \lambda_k v_k v_k^\top$ with $\lambda_k \geq 0$. Then $\bar{\mathcal{C}}(\pi, S) = \sum_k \lambda_k v_k^\top A_S^{-1} v_k$, a non-negative linear combination of the prediction-variance functions $\varphi_k(S) := v_k^\top A_S^{-1} v_k$. Each φ_k is monotone non-increasing and supermodular as a set function in S (Das and Kempe, 2008, Lemma 2.4), hence each $-\varphi_k$ is monotone non-decreasing and submodular, and $g = \sum_k \lambda_k (\varphi_k(S_0) - \varphi_k(S))$ inherits the same properties. $\square$

The submodular gain g has a closed-form marginal — the Sherman–Morrison expression (C.5) — which makes the greedy update cheap to evaluate:

$$g(S \cup \{i\}) - g(S) = \frac{\phi_i^\top A_S^{-1} \mathbf{A}_\pi A_S^{-1} \phi_i}{1 + \phi_i^\top A_S^{-1} \phi_i}.$$

Algorithm 6 Greedy Subset Selection for the LSTD Term

- 1: **Input:** feature pool $\{\phi_i\}_{i=1}^N$; target covariance $\mathbf{A}_\pi$; cardinality budget m .
 - 2: **Initialize:** anchor S_0 of size d chosen so that $A_{S_0} \succ 0$ (e.g., d greedy steps of $\max_i \det(A_{S \cup \{i\}})$ from $\emptyset$).
 - 3: $S \leftarrow S_0$.
 - 4: **for** $t = |S_0| + 1, \dots, m$ **do**
 - 5: Compute the marginal $g(S \cup \{i\}) - g(S)$ for each $i \notin S$ via the closed-form ratio above.
 - 6: $i^* \leftarrow \arg \max_{i \notin S} \frac{\phi_i^\top A_S^{-1} \mathbf{A}_\pi A_S^{-1} \phi_i}{1 + \phi_i^\top A_S^{-1} \phi_i}$.
 - 7: $S \leftarrow S \cup \{i^*\}$; update A_S^{-1} via Sherman–Morrison.
 - 8: **end for**
 - 9: **Output:** $\hat{S} = S$, $|\hat{S}| = m$.
-

Proposition 1 (($1 - 1/e$) Greedy Approximation). *Let $\hat{S}$ be the output of Algorithm 6 with budget m , and let $S_m^* \in \arg \min_{|S|=m, S \supseteq S_0} \bar{\mathcal{C}}(\pi, S)$. Then*

$$\bar{\mathcal{C}}(\pi, \hat{S}) - \bar{\mathcal{C}}(\pi, S_m^*) \leq \frac{1}{e} (\bar{\mathcal{C}}(\pi, S_0) - \bar{\mathcal{C}}(\pi, S_m^*)). \quad (\text{C.8})$$

Equivalently, the algorithm captures at least a $(1 - 1/e)$ fraction of the optimal gain $g(S_m^) = \bar{\mathcal{C}}(\pi, S_0) - \bar{\mathcal{C}}(\pi, S_m^*)$.*

Proof. The function g is non-negative, monotone, and submodular by Lemma 9, so cardinality-constrained submodular maximization with greedy enjoys the $(1 - 1/e)$ -approximation of Nemhauser et al. (1978, Theorem 4.1): $g(\hat{S}) \geq (1 - 1/e) g(S_m^*)$. Rearranging gives (C.8). $\square$

Combining Proposition 1 with the decomposition (C.7) yields a polynomial-time algorithm for (C.4): run Algorithm 6 for each candidate m in an outer 1-D grid, evaluate $(1 + \frac{m/K}{1-\gamma}) \bar{\mathcal{C}}(\pi, \hat{S}_m)$, and return the minimizer. Crucially, the algorithm runs whenever $N \geq d$ — no $N \gtrsim d/\varepsilon^2$ requirement — so the partial-coverage regime in which the offline buffer is only modestly larger than d remains fully addressable.

Continuous relaxation (heuristic alternative). Replacing the indicator $\mathbf{1}\{i \in S\}$ by a weight $w_i \geq 0$ with $\sum_i w_i = m$, the LSTD term of (C.4) becomes the classical *L-optimal experimental design* objective $\text{tr}(\mathbf{A}_\pi(\sum_i w_i \phi_i \phi_i^\top)^{-1})$, which is convex in w (Pukelsheim, 2006; Bhatia, 1997). The Kiefer–Wolfowitz equivalence theorem (Kiefer and Wolfowitz, 1960) and the Fedorov–Wynn multiplicative-weights iteration (Atwood, 1973) provide a fast first cut at the optimal continuous weights w^* ; thresholding w^* at its top- m support gives a heuristic subset. This is a relaxation only — the indicator constraint is dropped, and the trade-off factor $|S|/K$ is reintroduced post hoc. It is useful as a warm-start for Algorithm 6 but does not on its own come with the $(1 - 1/e)$ guarantee of Proposition 1.

Structural compressibility via spectral sparsification. A complementary, purely structural statement is that the buffer itself admits a small sub-sample preserving its entire spectrum. The Batson–Spielman–Srivastava sparsifier (Batson et al., 2014) and the column-subset-selection algorithm of Avron and Boutsidis (2013) construct such an object in polynomial time.

Theorem C.2 (Spectral Sparsifier of the Buffer). *Fix $\varepsilon \in (0, 1/2)$ and assume $A_{[N]} \succ 0$. There is a deterministic, polynomial-time algorithm (Batson et al., 2014; Avron and Boutsidis, 2013) that returns a subset $S \subseteq [N]$ of size*

$$|S| = O(d/\varepsilon^2)$$

such that the unweighted feature covariance $A_S = \sum_{i \in S} \phi_i \phi_i^\top$ satisfies the spectral approximation

$$(1 - \varepsilon) A_{[N]} \preceq A_S \preceq (1 + \varepsilon) A_{[N]},$$

and consequently $\bar{\mathcal{C}}(\pi, S) \leq (1 - \varepsilon)^{-1} \bar{\mathcal{C}}(\pi, [N])$.

Remark 6 (BSS is a structural guarantee, not a S^* solver). *Theorem C.2 is a compressibility statement: it demonstrates that a sample-rich buffer can be losslessly summarized by $O(d/\varepsilon^2)$ entries, but it does not solve (C.4). Three caveats apply in the partial-coverage regime relevant to offline RL. First, the size $O(d/\varepsilon^2)$ is set by the worst-case information-theoretic lower bound for spectral approximation, not by the trade-off objective; the actual $|S|^*$ for (C.4) can be either smaller (when the buffer contains substantial redundancy) or unattainable (when N is itself below $O(d/\varepsilon^2)$). Second, the bound compares $\bar{\mathcal{C}}(\pi, S)$ only against the full-buffer value $\bar{\mathcal{C}}(\pi, [N])$, not against the cardinality-restricted optimum $\min_{|S|=m} \bar{\mathcal{C}}(\pi, S)$; spectral sparsification is target-agnostic and does not exploit the geometry of $\mathbf{A}_\pi$. Third, when $N \lesssim d/\varepsilon^2$, the theorem becomes vacuous and offers no actionable subset. We therefore treat C.2 as a backup structural result; Algorithm 6 is the practical S^* solver, and it places no lower-bound requirement on N beyond $N \geq d$.*

Putting it together. Algorithm 6 delivers, for each candidate m , a subset $\hat{S}_m$ whose LSTD term is within a $(1 - 1/e)$ multiplicative factor of the cardinality-constrained optimum (Proposition 1). Plugging into Theorem C.1 reduces the full selection problem (C.4) to a 1-D scalar trade-off

$$m^* \in \arg \min_{m \in [d, N]} \left(1 + \frac{m/K}{1-\gamma}\right) \bar{\mathcal{C}}(\pi, \hat{S}_m),$$

which is solved by a single pass over the candidate grid. The total work is dominated by $O(N \cdot m^* \cdot d^2)$ for the greedy inner loop with Sherman–Morrison updates, independent of ε .

C.7 Continuous Sampling Distributions for Scalable Selection

The discrete solver of §C.6 returns a hard subset $\hat{S}$ and feeds Algorithm 2 the uniform measure $\mu_{\hat{S}}$. In the actor loop of Algorithm 3, however, the features $\phi_i = \phi^{\bar{\pi}^{(k)}}(s_i, a_i)$ change with every reference policy $\bar{\pi}^{(k)}$, so the cardinality-constrained greedy and its 1-D scan would have to be *re-solved from scratch at each epoch*; moreover, data collection and selection are typically asynchronous. Both considerations motivate replacing the hard subset by a *sampling distribution* over the buffer that can be refreshed incrementally rather than recomputed combinatorially. This subsection develops the corresponding continuous formulation; unlike the heuristic continuous relaxation noted in §C.6 (which keeps only the LSTD term and reintroduces the reuse penalty post hoc), it carries *both* layers of Theorem C.1 through exactly.

From subsets to sampling distributions. Let $\Delta_N := \{p \in \mathbb{R}_{\geq 0}^N : \sum_{i=1}^N p_i = 1\}$ be the probability simplex over the buffer indices. A *sampling distribution* $p \in \Delta_N$ generalizes a subset: drawing each inner-loop transition of Algorithm 2 i.i.d. from p (i.e. $\mu \leftarrow p$) recovers $\mu_S = \text{Unif}(S)$ exactly when p is uniform on S . The two covariance objects that drive the analysis are the *normalized design* and the *weight-squared covariance*

$$\Sigma_p := \sum_{i=1}^N p_i \phi_i \phi_i^\top, \quad M_p := \sum_{i=1}^N p_i^2 \phi_i \phi_i^\top.$$

With i.i.d. draws from p and $K \rightarrow \infty$ inner TD steps, the inner loop converges to the p -weighted truncated Bellman fixed point $\xi_p^* = \Sigma_p^{-1} b_p(\xi_p^*)$, the direct analogue of (C.2) with $A_S \leftarrow \Sigma_p$ and the summation in b_S reweighted by p_i (normalization cancels, so ξ_p^* reduces to ξ_S^* at $p = \text{Unif}(S)$).

The continuous trade-off. Tracing the two-layer decomposition of §C.3 through i.i.d. sampling from p replaces each unnormalized covariance A_S by the pair (Σ_p, M_p) . The two layers now arise from two *distinct* sources of stochasticity: the LSTD layer is the estimation variance of the weighted normal equation under the original buffer’s reward/transition noise (entering with weight p_i , so its contribution to the sandwich is proportional to p_i^2), whereas the TD-to-LSTD layer is the variance of finite- K stochastic approximation under i.i.d. resampling of buffer indices from p (entering with weight p_i).

The decomposition mirrors that of §C.3, with two continuous lemmas paralleling Lemmas 7 and 8. Throughout the continuous LSTD bound we condition on the buffer features $\{\phi_i\}$ and assume the sampling weights p are *feature-measurable*, i.e. they may depend on $\{\phi_i\}$, $\mathbf{A}_\pi$, and the policy π , but not on the realized Bellman residuals δ_i . This is the natural setting for the selection program (C.10), whose objective uses only Σ_p , M_p , and the leverage scores $\{\ell_i(p)\}$ — all feature-measurable.

Lemma 10 (Continuous LSTD Approximation Error). *Let Assumptions 2 and 3 hold, suppose $\Sigma_p \succ 0$, and assume the feature-measurability condition above. The p -weighted LSTD fixed point ξ_p^* satisfies*

$$\|\xi_p^* - \xi^*\|_{\mathbf{A}_\pi}^2 \lesssim (1 - \gamma)^{-3} \text{tr}(\mathbf{A}_\pi \Sigma_p^{-1} M_p \Sigma_p^{-1}) + \mathcal{E}_{\text{approx}}^2.$$

Proof sketch. Writing the per-sample Bellman targets at the fixed point as $y_i = \phi_i^\top \xi^* + \delta_i$ with δ_i a zero-mean residual (under Bellman completeness, modulo $\mathcal{E}_{\text{approx}}$) of variance bounded by the squared value range $(1 - \gamma)^{-2}$, the weighted fixed-point relation $\Sigma_p \xi_p^* = b_p$ gives $\xi_p^* - \xi^* = \Sigma_p^{-1} \sum_i p_i \phi_i \delta_i + O(\mathcal{E}_{\text{approx}})$. The Horvitz–Thompson-type noise $\sum_i p_i \phi_i \delta_i$ has covariance $\preceq (1 - \gamma)^{-2} M_p$, so $\mathbb{E} \|\xi_p^* - \xi^*\|_{\mathbf{A}_\pi}^2 \leq (1 - \gamma)^{-2} \text{tr}(\mathbf{A}_\pi \Sigma_p^{-1} M_p \Sigma_p^{-1}) + \mathcal{E}_{\text{approx}}^2$. An additional $(1 - \gamma)^{-1}$ from the projection/contraction conditioning of the truncated Bellman operator, absorbed by the same offline-LSTD machinery as Lemma 7 [yx:cite, same source as Lemma 7: duan2020minimax / min2021variance / xiong2022nearly], yields the stated $(1 - \gamma)^{-3}$ form. At $p = \text{Unif}(S)$, $\Sigma_p^{-1} M_p \Sigma_p^{-1} = A_S^{-1}$, recovering Lemma 7 exactly. $\square$

Lemma 11 (Continuous TD-to-LSTD Optimization Gap). *Under the same hypotheses, if Algorithm 2 is run on $\mu \leftarrow p$ with K inner-loop TD steps and target-network iterations T taken to convergence, the iterate $\hat{\xi}_p^{\text{TD}}$ satisfies*

$$\|\hat{\xi}_p^{\text{TD}} - \xi_p^*\|_{\mathbf{A}_\pi}^2 \lesssim (1 - \gamma)^{-4} \frac{1}{K} \text{tr}(\mathbf{A}_\pi \Sigma_p^{-1}).$$

Proof sketch. The inner loop is stochastic approximation toward the p -weighted LSTD fixed point with i.i.d. draws from p . Linearising around ξ_p^* gives the gradient-noise covariance $G_p = \sum_i p_i \rho_i^2 \phi_i \phi_i^\top$ where

$\rho_i := y_i - \phi_i^\top \xi_p^*$. The truncation bound on the values forces $\rho_i^2 \leq (1 - \gamma)^{-2}$, hence $G_p \preceq (1 - \gamma)^{-2} \Sigma_p$. The standard SA rate for K i.i.d. steps with conditioning Σ_p yields a finite-iteration variance of order $\frac{1}{K} \Sigma_p^{-1} G_p \Sigma_p^{-1} \preceq \frac{(1-\gamma)^{-2}}{K} \Sigma_p^{-1}$, so that $\mathbb{E} \|\hat{\xi}_p^{\text{TD}} - \xi_p^*\|_{\mathbf{A}_\pi}^2 \lesssim \frac{(1-\gamma)^{-2}}{K} \text{tr}(\mathbf{A}_\pi \Sigma_p^{-1})$. A further $(1 - \gamma)^{-2}$ from the truncated-Bellman contraction (matching Lemma 8's conditioning) gives the stated $(1 - \gamma)^{-4}$. At $p = \text{Unif}(S)$, $\text{tr}(\mathbf{A}_\pi \Sigma_p^{-1}) = |S| \bar{\mathcal{C}}(\pi, S)$, recovering Lemma 8 exactly. [yx: Inherits the same [yx:] caveat as Lemma 8: the $(1 - \gamma)^{-4} \cdot |S|/K \cdot \bar{\mathcal{C}}$ form is the worst-case bounding of $\bar{\rho}^2 \leq (1 - \gamma)^{-2}$; the data-dependent rate is $(1 - \gamma)^{-2} \bar{\rho}^2 \cdot \frac{1}{K} \text{tr}(\mathbf{A}_\pi \Sigma_p^{-1})$.] $\square$

Theorem C.3 (Continuous Selection-Aware Total Error). *Let Assumptions 1–3 hold and suppose $\Sigma_p \succ 0$. The output $\hat{\xi}_p^{\text{TD}}$ of Algorithm 2 run on $\mu \leftarrow p$ satisfies*

$$\|\hat{\xi}_p^{\text{TD}} - \xi^*\|_{\mathbf{A}_\pi}^2 \lesssim \underbrace{(1 - \gamma)^{-3} \text{tr}(\mathbf{A}_\pi \Sigma_p^{-1} M_p \Sigma_p^{-1})}_{\text{LSTD layer (Lemma 10)}} + \underbrace{(1 - \gamma)^{-4} \frac{1}{K} \text{tr}(\mathbf{A}_\pi \Sigma_p^{-1})}_{\text{TD-to-LSTD layer (Lemma 11)}} + \mathcal{E}_{\text{approx}}^2. \quad (\text{C.9})$$

Proof. Combine Lemmas 10 and 11 via the weak triangle inequality $\|\hat{\xi}_p^{\text{TD}} - \xi^*\|_{\mathbf{A}_\pi}^2 \leq 2\|\hat{\xi}_p^{\text{TD}} - \xi_p^*\|_{\mathbf{A}_\pi}^2 + 2\|\xi_p^* - \xi^*\|_{\mathbf{A}_\pi}^2$, and absorb the constant 2 into the $\lesssim$ notation, exactly as in the proof of Theorem C.1. $\square$

With K fixed, factoring out $(1 - \gamma)^{-3}$ yields the *continuous selection program*

$$p^* \in \arg \min_{p \in \Delta_N} F(p), \quad F(p) := \text{tr}(\mathbf{A}_\pi \Sigma_p^{-1} M_p \Sigma_p^{-1}) + \beta \text{tr}(\mathbf{A}_\pi \Sigma_p^{-1}), \quad \beta := \frac{1}{K(1-\gamma)}. \quad (\text{C.10})$$

The single knob β encodes the trade-off: $\beta \rightarrow 0$ (large K) leaves only the LSTD variance, $\beta \rightarrow \infty$ (small K) leaves only the L-optimal coverage term.

A compact leverage form. Both terms in (C.10) are governed by the same *target-aligned leverage* of sample i at the current distribution,

$$\ell_i(p) := \phi_i^\top \Sigma_p^{-1} \mathbf{A}_\pi \Sigma_p^{-1} \phi_i, \quad (\text{C.11})$$

the continuous counterpart of the Sherman–Morrison numerator in (C.5). Indeed $\text{tr}(\mathbf{A}_\pi \Sigma_p^{-1}) = \sum_i p_i \ell_i(p)$ and $\text{tr}(\mathbf{A}_\pi \Sigma_p^{-1} M_p \Sigma_p^{-1}) = \sum_i p_i^2 \ell_i(p)$, so the objective collapses to

$$F(p) = \sum_{i=1}^N p_i (p_i + \beta) \ell_i(p).$$

The L-optimal layer weights leverage linearly in p_i (it rewards *equalizing* leverage across the support, the Kiefer–Wolfowitz condition), whereas the LSTD layer weights it quadratically (it *penalizes concentrating* mass on any single sample), and their balance is set by β .

Remark 7 (Exact reduction to the discrete trade-off). *At $p = \text{Unif}(S)$ one has $\Sigma_p = \frac{1}{|S|} A_S$ and $M_p = \frac{1}{|S|^2} A_S$, whence $\Sigma_p^{-1} M_p \Sigma_p^{-1} = A_S^{-1}$ and $\text{tr}(\mathbf{A}_\pi \Sigma_p^{-1}) = |S| \bar{\mathcal{C}}(\pi, S)$. Thus (C.9) reduces to Theorem C.1 and (C.10) to the factored selection problem $(1 + \frac{|S|/K}{1-\gamma}) \bar{\mathcal{C}}(\pi, S)$ of (C.4). The continuous program is therefore a strict relaxation: it optimizes over all of Δ_N , with the discrete solutions sitting at the uniform-on-support vertices of each face.*

Remark 8 (Data-dependent rates). *Tracking the data-dependent variances rather than bounding them by the worst-case value range, the proofs of Lemmas 10 and 11 give the sharper bound*

$$\|\hat{\xi}_p^{\text{TD}} - \xi^*\|_{\mathbf{A}_\pi}^2 \lesssim (1 - \gamma)^{-1} \text{tr}(\mathbf{A}_\pi \Sigma_p^{-1} M_{p,\sigma} \Sigma_p^{-1}) + (1 - \gamma)^{-2} \frac{1}{K} \text{tr}(\mathbf{A}_\pi \Sigma_p^{-1} \Omega_p^{\text{TD}} \Sigma_p^{-1}) + \mathcal{E}_{\text{approx}}^2,$$

where $M_{p,\sigma} := \sum_i p_i^2 \sigma_i^2 \phi_i \phi_i^\top$ is the Horvitz–Thompson covariance of the buffer-noise contribution and $\Omega_p^{\text{TD}} := \sum_i p_i \rho_i^2 \phi_i \phi_i^\top$ is the per-step sampling covariance of the TD update at ξ_p^* . The bounded-range substitutions $\sigma_i^2, \rho_i^2 \leq (1 - \gamma)^{-2}$ give $M_{p,\sigma} \preceq (1 - \gamma)^{-2} M_p$ and $\Omega_p^{\text{TD}} \preceq (1 - \gamma)^{-2} \Sigma_p$, recovering the worst-case form (C.9).

Proposition 2 (The continuous program is non-convex). *The objective F in (C.10) is not convex on Δ_N . Concretely, take $d = 2$, $N = 3$, $\mathbf{A}_\pi = I_2$, and unit-norm features $\phi_1 = e_1$, $\phi_2 = e_2$, $\phi_3 = (e_1 + e_2)/\sqrt{2}$. With $p = (\frac{2}{5}, \frac{1}{2}, \frac{1}{10})$, $q = (\frac{1}{2}, \frac{2}{5}, \frac{1}{10})$, and $r = (p + q)/2$, the LSTD-layer functional $G(p) := \text{tr}(\mathbf{A}_\pi \Sigma_p^{-1} M_p \Sigma_p^{-1})$ satisfies*

$$G(p) = G(q) = \frac{4084}{2401}, \quad G(r) = \frac{206}{121} > \frac{1}{2}(G(p) + G(q)),$$

strictly violating midpoint convexity of G . Combined with the coverage layer $H(p) := \text{tr}(\mathbf{A}_\pi \Sigma_p^{-1})$ (which is convex), the full objective $F = G + \beta H$ is non-convex along this chord for every $\beta \in [0, \beta_\star)$ with $\beta_\star = \frac{1989}{53900} \approx 0.037$, i.e. whenever $K(1 - \gamma) \gtrsim 27$.

Proof. Direct computation: $\Sigma_p = \frac{1}{20} \begin{pmatrix} 9 & 1 \\ 1 & 11 \end{pmatrix}$, $\Sigma_p^{-1} = \frac{10}{49} \begin{pmatrix} 11 & -1 \\ -1 & 9 \end{pmatrix}$, $M_p = \frac{1}{200} \begin{pmatrix} 33 & 1 \\ 1 & 51 \end{pmatrix}$, hence $G(p) = \frac{4084}{2401}$, and $G(q) = G(p)$ by the symmetry $p \leftrightarrow q$, $e_1 \leftrightarrow e_2$. At $r = (\frac{9}{20}, \frac{9}{20}, \frac{1}{10})$, $\Sigma_r^{-1} = \frac{20}{99} \begin{pmatrix} 10 & -1 \\ -1 & 10 \end{pmatrix}$, $M_r = \frac{1}{400} \begin{pmatrix} 83 & 2 \\ 2 & 83 \end{pmatrix}$, yielding $G(r) = \frac{206}{121}$. For the coverage term $H(p) = \frac{200}{49} = H(q)$ and $H(r) = \frac{400}{99}$, so $H(r) - \frac{1}{2}(H(p) + H(q)) = -\frac{200}{4851} < 0$ (consistent with convexity of H). Hence $F(r) - \frac{1}{2}(F(p) + F(q)) = \frac{442}{290521} - \beta \cdot \frac{200}{4851}$, which is strictly positive iff $\beta < \frac{1989}{53900}$. $\square$

Lemma 12 (Exact LSTD-layer floor). *Let $V := [\phi_1, \dots, \phi_N] \in \mathbb{R}^{d \times N}$ and $A_B := VV^\top = \sum_{i=1}^N \phi_i \phi_i^\top$, the full-buffer feature Gram. Assume $A_B \succ 0$. For every p in the relative interior of Δ_N ,*

$$G(p) := \text{tr}(\mathbf{A}_\pi \Sigma_p^{-1} M_p \Sigma_p^{-1}) \geq \text{tr}(\mathbf{A}_\pi A_B^{-1}), \quad (\text{C.12})$$

with equality at the full-buffer uniform distribution $u := (1/N, \dots, 1/N)$. Equality holds, more generally, iff $\text{Row}(V) = \text{Row}(VD_p)$ as subspaces of $\mathbb{R}^N$, i.e. iff $\text{Row}(V)$ is invariant under right-multiplication by D_p . For $d < N$ and generic V (so that no proper non-scalar positive diagonal preserves $\text{Row}(V)$), this forces $p = u$ and u is the unique interior minimizer of G . (When $d = N$, V is square and invertible, $\text{Row}(V) = \mathbb{R}^N$ is trivially D_p -invariant, and G is constant on the interior.)

Proof. Let $W := VD_p$ with $D_p = \text{diag}(p)$, so that $\Sigma_p = WV^\top$ and $M_p = WW^\top$. Then

$$C_p := \Sigma_p M_p^{-1} \Sigma_p = V \underbrace{W^\top (WW^\top)^{-1} W}_{=: \Pi_W} V^\top,$$

where Π_W is the orthogonal projection in $\mathbb{R}^N$ onto $\text{Row}(W)$. Since $\Pi_W \preceq I_N$, $A_B - C_p = V(I_N - \Pi_W)V^\top \succeq 0$, hence $C_p \preceq A_B$, and (both being positive definite) taking inverses flips the inequality: $\Sigma_p^{-1} M_p \Sigma_p^{-1} = C_p^{-1} \succeq A_B^{-1}$. Taking the $\mathbf{A}_\pi$ -weighted trace yields (C.12). At $p = u$, $D_u = \frac{1}{N} I_N$ gives $\text{Row}(W) = \text{Row}(VD_u) = \text{Row}(V)$, so $V(I_N - \Pi_W) = 0$ and equality holds. For generic V , equality forces $\text{Row}(V)$ to be D_p -invariant; since a positive diagonal preserves a non-coordinate subspace only when its entries are equal, this implies $p = u$. $\square$

Remark 9 (Endpoint structure of the continuous trade-off). *Lemma 12 and the convexity of the coverage layer pin down the two endpoints of the program (C.10). As $\beta \rightarrow 0$ (large- K / variance-dominated), $F \rightarrow G$ and its unique interior minimizer is the full-buffer uniform distribution u , with optimal value $\text{tr}(\mathbf{A}_\pi A_B^{-1})$ — the continuous counterpart of "use the entire buffer" in the discrete trade-off. As $\beta \rightarrow \infty$ (small- K / coverage-dominated), $F \rightarrow \beta H$ and its unique minimizer is the $\mathbf{L}$ -optimal design on $\{\phi_i\}$ (Kiefer and Wolfowitz, 1960; Atwood, 1973), generally sparse. The non-convex regime identified in Proposition 2 sits in the intermediate range of β , away from both endpoints. The theory anchors the two endpoints: for sufficiently small β , the global minimizer lies in any prescribed neighborhood of u (Lemma 13), while for large β the coverage layer drives the minimizer toward the $\mathbf{L}$ -optimal design. Empirically, in the synthetic study of §C.9, the optimizer traces a smooth path between these two endpoints (the continuous analogue of the discrete $|S|^\star \sim K$ curve); a uniform analytic statement of this continuity remains open.*

Lemma 13 (Small- β branch around the uniform endpoint). *Assume the hypotheses of Lemma 12 (so u is the unique interior minimizer of G) and $\nabla_T^2 G(u) \succ 0$ on the simplex tangent $T_\Delta := \{v : \mathbf{1}^\top v = 0\}$.*

1. Global minimum localisation. *For every neighborhood $B_\epsilon(u)$ of u there exist $\delta_\epsilon > 0$ and $\beta_0 = \delta_\epsilon / H(u) > 0$ such that for all $\beta \in [0, \beta_0)$ the global minimizer of $F = G + \beta H$ on the interior of Δ_N lies in $B_\epsilon(u)$.*

2. Smooth stationary branch. *There exists $\bar{\beta} > 0$ and a unique smooth branch $\beta \mapsto p_\beta$ of KKT points of (C.10) on $[0, \bar{\beta})$ with $p_0 = u$ and*

$$p_\beta = u - \beta (\nabla_T^2 G(u))^{-1} P_T \nabla H(u) + O(\beta^2), \quad P_T := I_N - \frac{1}{N} \mathbf{1} \mathbf{1}^\top. \quad (\text{C.13})$$

Proof. (i) By Lemma 12, $G(p) > G(u)$ for every interior $p \neq u$, with $G(p) \rightarrow +\infty$ as Σ_p becomes singular. Hence $\delta_\epsilon := \inf\{G(p) - G(u) : p \in \text{int}(\Delta_N) \setminus B_\epsilon(u)\} > 0$. For $\beta < \delta_\epsilon/H(u)$ and $p \notin B_\epsilon(u)$ with $\Sigma_p \succ 0$, $H(p) \geq 0$ gives $F(p) - F(u) \geq \delta_\epsilon - \beta H(u) > 0$, so the global minimizer on the interior lies in $B_\epsilon(u)$.

(ii) The simplex KKT system reads $\nabla F(p) - \tau \mathbf{1} = 0$, $\mathbf{1}^\top p = 1$. At $(\beta, p, \tau) = (0, u, 0)$ it is satisfied, and its Jacobian in (p, τ) is the bordered matrix $\begin{pmatrix} \nabla^2 G(u) & -\mathbf{1} \\ \mathbf{1}^\top & 0 \end{pmatrix}$, which is invertible because $\nabla_T^2 G(u) \succ 0$ on T_Δ . The implicit function theorem yields a unique smooth branch p_β ; linearising $\nabla F(p_\beta) = \tau_\beta \mathbf{1}$ in β at 0 and projecting onto T_Δ gives (C.13). $\square$

Lemma 14 (Gradient and first-order identities). *For p in the interior of Δ_N with $\Sigma_p \succ 0$, set $\ell_i(p) := \phi_i^\top \Sigma_p^{-1} \mathbf{A}_\pi \Sigma_p^{-1} \phi_i$ (as in (C.11)) and $h_i(p) := \phi_i^\top \Sigma_p^{-1} \mathbf{A}_\pi \Sigma_p^{-1} M_p \Sigma_p^{-1} \phi_i$. The partial derivatives of the two layers and of F are*

$$\nabla_i G(p) = 2p_i \ell_i(p) - 2h_i(p), \quad \nabla_i H(p) = -\ell_i(p), \quad \nabla_i F(p) = (2p_i - \beta) \ell_i(p) - 2h_i(p). \quad (\text{C.14})$$

Moreover the following three identities hold:

$$\sum_{i=1}^N p_i \ell_i(p) = H(p), \quad \sum_{i=1}^N p_i^2 \ell_i(p) = G(p), \quad \sum_{i=1}^N p_i h_i(p) = G(p), \quad (\text{C.15})$$

and consequently the radial inner product simplifies to

$$\langle p, \nabla F(p) \rangle = -\beta H(p). \quad (\text{C.16})$$

Proof. Differentiate via $\partial \Sigma_p / \partial p_i = \phi_i \phi_i^\top$, $\partial M_p / \partial p_i = 2p_i \phi_i \phi_i^\top$, and $\partial(\Sigma_p^{-1}) / \partial p_i = -\Sigma_p^{-1} \phi_i \phi_i^\top \Sigma_p^{-1}$. For $G(p) = \text{tr}(\mathbf{A}_\pi \Sigma_p^{-1} M_p \Sigma_p^{-1})$, the M_p -derivative contributes $2p_i \phi_i^\top \Sigma_p^{-1} \mathbf{A}_\pi \Sigma_p^{-1} \phi_i = 2p_i \ell_i$, and the two symmetric Σ_p^{-1} -derivatives combine (by cyclicity) into $-2 \phi_i^\top \Sigma_p^{-1} \mathbf{A}_\pi \Sigma_p^{-1} M_p \Sigma_p^{-1} \phi_i = -2h_i$, giving $\nabla_i G = 2p_i \ell_i - 2h_i$. For $H(p) = \text{tr}(\mathbf{A}_\pi \Sigma_p^{-1})$, $\nabla_i H = -\phi_i^\top \Sigma_p^{-1} \mathbf{A}_\pi \Sigma_p^{-1} \phi_i = -\ell_i$, proving (C.14). The identities (C.15) follow by writing each sum as a trace: $\sum_i p_i \ell_i = \text{tr}(\Sigma_p^{-1} \mathbf{A}_\pi \Sigma_p^{-1} \Sigma_p) = H$; $\sum_i p_i^2 \ell_i = \text{tr}(\Sigma_p^{-1} \mathbf{A}_\pi \Sigma_p^{-1} M_p) = G$; and $\sum_i p_i h_i = \text{tr}(\Sigma_p^{-1} \mathbf{A}_\pi \Sigma_p^{-1} M_p \Sigma_p^{-1} \Sigma_p) = G$. Combining gives $\langle p, \nabla F \rangle = 2G - \beta H - 2G = -\beta H$. $\square$

Proposition 3 (Frank–Wolfe gap and KKT characterisation). *For $p \in \Delta_N$ with $\Sigma_p \succ 0$, define the vertex-descent score at each $j \in [N]$ by*

$$E_j^{\text{all}}(p) := -DF(p)[e_j - p] = \beta (\ell_j(p) - H(p)) + 2h_j(p) - 2p_j \ell_j(p), \quad (\text{C.17})$$

and the Frank–Wolfe gap of F by

$$g_{\text{FW}}(p) := \max_{j \in [N]} E_j^{\text{all}}(p) \stackrel{(\text{C.16})}{=} -\beta H(p) - \min_{j \in [N]} \nabla_j F(p).$$

Then $g_{\text{FW}}(p) \geq 0$ always, and $g_{\text{FW}}(p) = 0$ if and only if p is a first-order KKT point of (C.10), with Lagrange multiplier $\tau = -\beta H(p)$. In particular, $g_{\text{FW}}(p) > 0$ implies p is not a local minimum: the vertex e_{j^*} attaining the max yields a feasible descent direction $e_{j^*} - p$.

Proof. For any j , $DF(p)[e_j - p] = \nabla_j F(p) - \langle p, \nabla F(p) \rangle = \nabla_j F(p) + \beta H(p)$ by (C.16), so the two expressions for g_{FW} agree. Non-negativity: $\sum_j p_j (-DF(p)[e_j - p]) = -\langle p, \nabla F(p) \rangle - \beta H(p) = 0$ by (C.16) and the simplex constraint, so the maximum is $\geq$ the (zero) weighted mean. The simplex KKT condition is the existence of τ with $\nabla_i F(p) = \tau$ on $\text{supp}(p)$ and $\nabla_j F(p) \geq \tau$ off; multiplying by p_i and summing gives $\tau = \langle p, \nabla F \rangle = -\beta H(p)$, so KKT reads $\min_j \nabla_j F(p) \geq -\beta H(p)$, i.e. $g_{\text{FW}}(p) \leq 0$. Combined with the unconditional lower bound this is $g_{\text{FW}}(p) = 0$. $\square$

Proposition 4 (Boundary-escape certificate). *For $j \notin \text{supp}(p)$, the vertex-descent score $E_j^{\text{all}}(p)$ of (C.17) simplifies (since $p_j = 0$) to the off-support escape score*

$$E_j(p) := \beta (\ell_j(p) - H(p)) + 2h_j(p).$$

As a corollary of Proposition 3:

1. **Descent at the boundary.** *If $p_j = 0$ for some j with $E_j(p) > 0$, then $g_{\text{FW}}(p) \geq E_j(p) > 0$, hence $e_j - p$ is a feasible descent direction of F at p and p is not a local minimum.*
2. **Off-support KKT condition.** *If p is a first-order KKT point of (C.10) with support I , then $E_j(p) \leq 0$ for every $j \notin I$.*

Proof. For $j \notin \text{supp}(p)$, $E_j^{\text{all}}(p) = E_j(p)$ by (C.17), and both items follow from Proposition 3: (i) from $g_{\text{FW}}(p) \geq E_j(p)$ and the descent characterisation; (ii) from $g_{\text{FW}}(p) \leq 0$ at KKT points and $E_j(p) \leq g_{\text{FW}}(p) \leq 0$ off-support. $\square$

Lemma 15 (Hessian quadratic forms). *Assume $\mathbf{A}_\pi \succ 0$. Let $B := \Sigma_p^{-1}$ and, for a simplex-tangent direction $v \in T_\Delta := \{v \in \mathbb{R}^N : \mathbf{1}^\top v = 0\}$, write $S_v := \sum_i v_i \phi_i \phi_i^\top$, $R_v := 2 \sum_i p_i v_i \phi_i \phi_i^\top$ and $U_v := 2 \sum_i v_i^2 \phi_i \phi_i^\top$. The tangent quadratic forms of the two layer Hessians are*

$$Q_H(p; v) := v^\top \nabla^2 H(p) v = 2 \text{tr}(\mathbf{A}_\pi B S_v B S_v B), \quad (\text{C.18})$$

$$\begin{aligned} Q_G(p; v) := v^\top \nabla^2 G(p) v &= 2 \text{tr}(\mathbf{A}_\pi B S_v B S_v B M_p B) + 2 \text{tr}(\mathbf{A}_\pi B M_p B S_v B S_v B) \\ &\quad + 2 \text{tr}(\mathbf{A}_\pi B S_v B M_p B S_v B) - 2 \text{tr}(\mathbf{A}_\pi B S_v B R_v B) - 2 \text{tr}(\mathbf{A}_\pi B R_v B S_v B) + \text{tr}(\mathbf{A}_\pi B U_v B). \end{aligned} \quad (\text{C.19})$$

Moreover:

1. $Q_H(p; v) \geq 0$, with $Q_H(p; v) = 0$ if and only if $S_v = 0$.
2. On the design-null subspace $\{v \in T_\Delta : S_v = 0\}$,

$$Q_G(p; v) = 2 \sum_{i=1}^N v_i^2 \ell_i(p) \geq 0,$$

since $\ell_i(p) \geq 0$ when $\mathbf{A}_\pi \succeq 0$.

Proof. Apply the standard inverse-perturbation expansion $(\Sigma_p + tS_v)^{-1} = B - tBS_vB + t^2BS_vBS_vB + O(t^3)$ and $M_{p+tv} = M_p + tR_v + \frac{t^2}{2}U_v$ to $H(p+tv) = \text{tr}(\mathbf{A}_\pi B_t)$ and $G(p+tv) = \text{tr}(\mathbf{A}_\pi B_t M_t B_t)$, then extract the coefficient of t^2 via the product rule. For (C.18), writing $Y := B^{1/2}S_vB^{1/2}$ and $C := B^{1/2}\mathbf{A}_\pi B^{1/2} \succeq 0$, $Q_H = 2 \text{tr}(CY^2) = 2\|C^{1/2}Y\|_F^2 \geq 0$, with equality iff $Y = 0$, i.e. iff $S_v = 0$. When $S_v = 0$ all but the last term in (C.19) vanish, and the residual is $\text{tr}(\mathbf{A}_\pi B U_v B) = 2 \sum_i v_i^2 \phi_i^\top B \mathbf{A}_\pi B \phi_i = 2 \sum_i v_i^2 \ell_i(p)$. $\square$

Proposition 5 (Large- β Hessian-domination certificate). *For $\rho > 0$, define the well-conditioned region $\mathcal{D}_\rho := \{p \in \Delta_N : \Sigma_p \succeq \rho I_d\}$ and the threshold*

$$\beta_\star(\rho) := \sup_{\substack{p \in \mathcal{D}_\rho, v \in T_\Delta, \|v\|=1 \\ Q_H(p; v) > 0}} \frac{[-Q_G(p; v)]_+}{Q_H(p; v)}. \quad (\text{C.20})$$

Then for every $\beta \geq \beta_\star(\rho)$, the objective $F = G + \beta H$ satisfies $v^\top \nabla^2 F(p) v \geq 0$ for all $p \in \mathcal{D}_\rho$ and all tangent directions $v \in T_\Delta$; in particular F is convex on $\mathcal{D}_\rho$ along simplex-feasible directions.

Proof. For any $p \in \mathcal{D}_\rho$ and $v \in T_\Delta$, $v^\top \nabla^2 F(p) v = Q_G(p; v) + \beta Q_H(p; v)$ by linearity of the Hessian in β . If $Q_H(p; v) > 0$, the definition of $\beta_\star(\rho)$ gives $-Q_G(p; v) \leq \beta_\star(\rho) Q_H(p; v)$, whence $Q_G + \beta Q_H \geq (\beta - \beta_\star(\rho)) Q_H \geq 0$. If $Q_H(p; v) = 0$, Lemma 15.(i) forces $S_v = 0$, and Lemma 15.(ii) then gives $Q_G(p; v) = 2 \sum_i v_i^2 \ell_i(p) \geq 0$. $\square$

Proposition 6 (Conditional certificate for absence of spurious local minima). *Fix $\beta > 0$ and let $F^\star := \inf_{p \in \Delta_N} F(p)$. Suppose that for every $p \in \Delta_N$ with $\Sigma_p \succ 0$ and $F(p) > F^\star$, at least one of the following holds:*

(C1) $g_{\text{FW}}(p) > 0$;

(C2) $g_{\text{FW}}(p) = 0$ and $\lambda_{\min}^{\text{crit}}(p) < 0$,

where $\lambda_{\min}^{\text{crit}}(p) := \min_{v \in \mathcal{C}(p), \|v\|=1} v^\top \nabla^2 F(p) v$ on the critical cone $\mathcal{C}(p) := \{v : \mathbf{1}^\top v = 0, \langle \nabla F(p), v \rangle = 0, v_j \geq 0 \text{ for } p_j = 0\}$. Then F has no spurious local minima on Δ_N . Moreover, every non-global first-order KKT point of (C.10) admits a negative-curvature direction on the critical cone $\mathcal{C}(p)$; hence there are no non-global second-order KKT points.

Proof. Let p be non-global, $F(p) > F^*$. By hypothesis, p satisfies (C1) or (C2). If (C1) $g_{\text{FW}}(p) > 0$, then p is not a first-order KKT point (Proposition 3) and a fortiori not a local minimum. If (C2) $g_{\text{FW}}(p) = 0$ and $\lambda_{\min}^{\text{crit}}(p) < 0$, then p is a first-order KKT point with a negative-curvature direction on $\mathcal{C}(p)$, hence fails the second-order necessary condition for a local minimum (and is in particular not a second-order KKT point). $\square$

Remark 10 (Conjecture: benign continuous-selection landscape). (Geometry.) *For generic buffer features $\{\phi_i\}$, generic target covariance $\mathbf{A}_\pi \succ 0$, and any $\beta > 0$, every first-order KKT point of (C.10) is either globally optimal or has a negative-curvature direction on the critical cone $\mathcal{C}(p)$, i.e. the hypothesis of Proposition 6 holds.* (Algorithm.) *Consequently, projected gradient descent with random initialisation, or with arbitrarily small perturbations, converges to a global minimizer almost surely. The available pieces of the simplex are covered by complementary certificates: Lemma 13 (small- β branch around u), Proposition 5 (well-conditioned core $\mathcal{D}_\rho$), and Propositions 3 and 4 (Frank–Wolfe / boundary-escape on the sparse boundary). Assembling them into a single uniform statement is left open. Empirical support is reported in §C.9: in exp8, 160 projected-gradient runs across $\beta \in \{10, 1, 0.1, 0.01, 0.001\}$ and seven initialisation families yield no KKT-bad-local terminations.*

Solving the continuous program. The coverage layer $\text{tr}(\mathbf{A}_\pi \Sigma_p^{-1})$ is convex in p (operator convexity of $X \mapsto X^{-1}$ composed with the linear map $p \mapsto \Sigma_p$ (Bhatia, 1997)), and its minimizer is the $\mathbf{L}$ -optimal design, characterized by the Kiefer–Wolfowitz equivalence condition “ $\ell_i(p)$ equal on the support” (Kiefer and Wolfowitz, 1960) and computable by the Fedorov–Wynn multiplicative-weights update $p_i \leftarrow p_i \ell_i(p) / \sum_j p_j \ell_j(p)$ (Atwood, 1973). The LSTD layer $\text{tr}(\mathbf{A}_\pi \Sigma_p^{-1} M_p \Sigma_p^{-1})$ is the matrix-fractional variance of a weighted least-squares estimator and is *not* convex on all of Δ_N (Proposition 2; the same conclusion is reached by the tangent-Hessian probe in §C.9, which locates the non-convexity at near-boundary, low- n_{eff} distributions). The non-convexity of G is bounded, however: by Proposition 5, for every well-conditioned region $\mathcal{D}_\rho$ the (possibly infinite) threshold $\beta_\star(\rho)$ of (C.20) controls convexity — whenever $\beta \geq \beta_\star(\rho)$, $\nabla^2 F \succeq 0$ on $\mathcal{D}_\rho$ along simplex-feasible directions. The certificate is computable: on the synthetic buffer of §C.9, the sampled estimate of the supremum in (C.20) satisfies $\hat{\beta}_\star(\rho) \leq 0.025$ over the region $\rho \geq 10^{-2}$ reached by FW/PGD — well below the $\beta = 1$ at which the Hessian probes already certify PSD — with zero null-direction violations across all sampled probes (so the analytic null-safety of Lemma 15.(ii) is consistent with the data; the theorem itself remains a statement about the exact $\beta_\star(\rho)$, and the sampled estimate $\hat{\beta}_\star(\rho)$ does not by itself certify $\beta_\star(\rho) < \infty$). So convexity is certified on the well-conditioned core; non-convexity is confined to the sparse boundary regime $\rho \ll 1$, where Proposition 2 also bites. In the latter regime, however, the non-convexity is diagnosable as *boundary-escape* via the Frank–Wolfe gap $g_{\text{FW}}(p)$ of Proposition 3: any non-KKT p has $g_{\text{FW}}(p) > 0$ and the vertex direction $e_{j^\star} - p$ is descent; specialising to $p_j = 0$ recovers Proposition 4. Empirically (§C.9), among the tangent-Hessian-negative probes of exp6, $g_{\text{FW}}(p) > 0$ holds in all 79 such probes spanning $\beta \in \{0.1, 0.01, 0.001\}$, and the maximising vertex $j^\star$ lies among the lowest-mass coordinates (with $p_{j^\star} < 0.02$) in 77/79 cases — so the non-convexity carries an explicit boundary-pointing descent direction. In exp8, every projected-gradient terminal across 160 runs is either non-KKT (still moving along an FW descent) or KKT-global; no KKT-bad-local termination is observed.

The full assembly — small- β branch (Lemma 13), well-conditioned core (Proposition 5), and sparse-boundary escape (Propositions 3–4) — gives the conditional certificate Proposition 6: if every non-global $p \in \Delta_N$ satisfies either condition (C1) or (C2), then F has no spurious local minima. The empirical support for the hypothesis is uniform across our experiments (§C.9), but a uniform analytic verification remains open and is recorded as Conjecture 10. In all cases a practical solver runs the Fedorov–Wynn iteration as a warm start followed by projected gradient on F ; crucially, when the reference policy advances $\bar{\pi}^{(k)} \rightarrow \bar{\pi}^{(k+1)}$ and the features drift, the previous $p^{(k)}$ is a warm start for the next solve, giving the incremental, asynchronous refresh that the hard-subset solver cannot.

C.8 Total Sample Complexity for ϵ_g -Stationarity

The development so far fixes the target policy π and bounds the *per-evaluation* error of the functional critic on a selected subset. We now lift this to the full actor-critic loop of Algorithm 3 and read off the *total* number of TD samples that drives the non-convex policy-optimization guarantee of Theorem 3.2 to an ϵ_g -stationary point in average. The accounting is a product of two independent factors: the inner sample budget K that each critic evaluation needs to reach the floor, and the number of actor steps M over which evaluations are repeated. Selection acts on the first factor; the second multiplies it.

The bridge. In the main analysis the per-epoch critic error $\sqrt{\mathbb{E}[\|\xi^{(k)} - \xi^*\|_{\mathbf{A}_k}^2]}$ is the *single* quantity that propagates into the averaged-gradient bound (3.5) through the floor $\bar{\epsilon}$. Evaluating on the selected subset S replaces it by the selection-aware error of Theorem C.1, which we split into its K -independent and K -dependent parts:

$$\bar{\epsilon}_S^2 := \underbrace{(1-\gamma)^{-3} \bar{\mathcal{C}}(\pi, S) + \mathcal{E}_{\text{approx}}^2}_{K\text{-independent floor}} + \underbrace{(1-\gamma)^{-4} \frac{\mathcal{C}(\pi, \mu_S)}{K}}_{\text{TD-to-LSTD gap}}, \quad \mathcal{C}(\pi, \mu_S) := |S| \bar{\mathcal{C}}(\pi, S), \quad (\text{C.21})$$

where $\mathcal{C}(\pi, \mu_S)$ is the *normalized* coverage of Definition 1 and K is the inner-loop TD budget of Algorithm 2. Substituting $\bar{\epsilon}^2 \mapsto \bar{\epsilon}_S^2$ in (3.5) gives, for a suitable stepsize η ,

$$\frac{1}{M} \mathbb{E} \left[\sum_{k,t} \|\nabla J_\rho(\pi_t^{(k)})\|^2 \right] \lesssim \frac{\Delta J}{\eta M} + \bar{\epsilon}_S^2, \quad \Delta J := J_\rho^* - J_\rho(\pi^{(0)}). \quad (\text{C.22})$$

[yx:

In the current formulation, optimizing $MK \leq T$ setting requires future knowledge of $\pi^{(t)}$ sequence, which is un-available,

Option 1: let's formulate it as an dynamic, sequentially selection problem, where the goal is no-regret compare with the static benchmark know everything at the beginning.

Option 2: Let's make some structural assumption on how π changes(e.g., an upper bound on its leaving rate), then optimize such kind of heuristic objective.

If our budget is given as $MK = T$, then (C.22) can be transformed into

$$\min_{K \leq T, S} \quad \frac{K}{T} + (1-\gamma)^{-3} \bar{\mathcal{C}}(\pi, S) + \mathcal{E}_{\text{approx}}^2 + \underbrace{(1-\gamma)^{-4} \frac{\mathcal{C}(\pi, \mu_S)}{K}}_{\text{TD-to-LSTD gap}}.$$

Now the problem goes to

$$\min_S \quad (1-\gamma)^{-3} \bar{\mathcal{C}}(\pi, S) + \mathcal{E}_{\text{approx}}^2$$

draft:

$$K/T = (1-\gamma)^{-4} \mathcal{C}(\pi, \mu_S)/K \iff K = (1-\gamma)^{-2} \sqrt{TC(\pi, \mu_S)} \iff K/T = (1-\gamma)^{-2} \sqrt{\frac{\mathcal{C}(\pi, \mu_S)}{T}}.$$

This is, the optimal design goal now turns to

$$\min_S \quad (1-\gamma)^{-3} \bar{\mathcal{C}}(\pi, S) + (1-\gamma)^{-2} \sqrt{\frac{\mathcal{C}(\pi, \mu_S)}{T}}$$

]

[yx:

Difference between given K, M to attain minimum err and fix an err ϵ_g then finding minimum K with buffer design(suppose $M \asymp \epsilon_g^{-2}$ is fixed.):

For the second problem: Our goal is to optimize the hard constrained problem:

$$\min_S \mathcal{C}(\pi, \mu_S), \quad \text{s.t. } \bar{\mathcal{C}}(\pi, S) \leq \epsilon_g.$$

On the other hand, for the first problem, for a given M, K , our goal is to minimize the RHS of (C.22).]

Corollary 2 (Total Sample Complexity under Selection). *Let the assumptions of Theorems 3.2 and C.1 hold, and fix a target accuracy ϵ_g above the K -independent floor, i.e. $(1-\gamma)^{-3}\bar{\mathcal{C}}(\pi, S) + \mathcal{E}_{\text{approx}}^2 \lesssim \epsilon_g^2$. Run Algorithm 3 with every functional-critic evaluation drawn from the subset S . Then $\frac{1}{M}\mathbb{E}[\sum_{k,t} \|\nabla J_\rho(\pi_t^{(k)})\|^2] \leq \epsilon_g^2$ holds as soon as*

- **(per-evaluation budget)** each evaluation uses

$$K \gtrsim K_\star(S) := \frac{\mathcal{C}(\pi, \mu_S)}{(1-\gamma)^4 \epsilon_g^2},$$

which forces the TD-to-LSTD gap in (C.21) below $\frac{1}{2}\epsilon_g^2$, *[yx: and at the same time, we implicitly requires $\bar{\mathcal{C}}(\pi, S) \lesssim \epsilon_g^2$]* and

- **(optimization horizon)** the actor runs for $M \gtrsim \Delta J / (\eta \epsilon_g^2)$ steps, which forces the $\Delta J / (\eta M)$ term below $\frac{1}{2}\epsilon_g^2$.

Since each of the M actor steps triggers at most one critic evaluation, the total number of TD samples is the product of the two budgets,

$$N_{\text{tot}}(S) \leq \underbrace{M}_{\Theta(\epsilon_g^{-2})} \cdot \underbrace{K_\star(S)}_{\Theta(\epsilon_g^{-2})} \asymp \frac{\Delta J \mathcal{C}(\pi, \mu_S)}{\eta (1-\gamma)^4 \epsilon_g^4}. \quad (\text{C.23})$$

Proof. Switching Algorithm 2 to sample from μ_S changes the analysis only through the per-epoch critic error (3.3), which Theorem C.1 bounds by $\bar{\epsilon}_S^2$; this yields (C.22). The budget $K \gtrsim K_\star(S)$ makes the TD-to-LSTD gap term of (C.21) at most $\frac{1}{2}\epsilon_g^2$, and together with the floor hypothesis $\bar{\epsilon}_S^2 \lesssim \epsilon_g^2$, the horizon $M \gtrsim \Delta J / (\eta \epsilon_g^2)$ makes the optimization term at most $\frac{1}{2}\epsilon_g^2$. Repeating one evaluation of cost $K_\star(S)$ at each of the M steps gives $N_{\text{tot}} \leq M K_\star(S)$; substituting both budgets gives the closed form. $\square$

The rate $\Theta(\epsilon_g^{-4})$ is the canonical product of a $\Theta(\epsilon_g^{-2})$ non-convex optimization horizon and a $\Theta(\epsilon_g^{-2})$ per-evaluation estimation budget; the two ϵ_g^{-2} factors have *disjoint* origins and only the second sees the buffer through $\mathcal{C}(\pi, \mu_S)$.

Per-step vs. epoch accounting. Equation (C.23) charges one evaluation per actor step – the conservative upper bound matching the $\times M$ factor. The epoch schedule of Algorithm 3 reuses a single anchored critic across the T_k steps of epoch k , so the number of evaluations is the epoch count $\sum_k 1 \leq M$ and N_{tot} shrinks by the average epoch length; the ϵ_g^{-4} rate and the $\mathcal{C}(\pi, \mu_S)$ dependence are unchanged.

Remark 11 (Selection’s multiplicative saving in total samples). *Only the per-evaluation factor $K_\star(S) \propto \mathcal{C}(\pi, \mu_S)$ in (C.23) depends on the data, so selection rescales the entire $\Theta(\epsilon_g^{-4})$ budget through the normalized coverage. The naive baseline evaluates on the whole i.i.d. buffer, $S = [N]$, whose normalized coverage converges to the behavior constant, $\mathcal{C}(\pi, \mu_{[N]}) \rightarrow \mathcal{C}(\pi, \mu)$. Selection instead concentrates the normalized covariance $A_{S^\star} / |S^\star|$ along the $\mathbf{A}_\pi$ -relevant directions, dropping samples whose features are nearly $\mathbf{A}_\pi$ -orthogonal (the finite- K regime of §C.4); such samples inflate $\mathcal{C}(\pi, \mu)$ without aiding the target-aligned directions, so removing them yields $\mathcal{C}(\pi, \mu_{S^\star}) < \mathcal{C}(\pi, \mu)$. The total sample counts therefore obey*

$$\frac{N_{\text{tot}}(S^\star)}{N_{\text{tot}}([N])} \asymp \frac{\mathcal{C}(\pi, \mu_{S^\star})}{\mathcal{C}(\pi, \mu)} < 1,$$

a multiplicative saving that holds at every accuracy level ϵ_g : selection reaches the same ϵ_g -stationarity of the non-convex objective with strictly fewer total TD samples than the naive full-buffer-i.i.d. baseline, the gain being the ratio by which a target-aligned subset improves the behavior policy’s raw coverage constant.

C.9 Numerical Illustration: Purpose and Setup

We probe the continuous program (C.10) numerically. The study takes the bound (C.9) as given (it does *not* re-validate the constants, which is the role of Lemma 8); its purpose is to characterize the *geometry* of F and the quality of cheap solvers.

Questions.

1. **Non-triviality.** Does F admit an interior optimum p^* strictly better than both the uniform distribution and a degenerate concentration, and how does p^* reallocate mass across samples of differing target alignment?
2. **Tractability.** Distinguish (A) global convexity of F on Δ_N from (B) a merely *benign* non-convex landscape (all stationary points global / single basin). The former would yield a convex-program guarantee; the latter only an algorithmic one for projected gradient.
3. **Heuristic quality.** How close to the global optimum is the cheap leverage-based solver (Fedorov–Wynn warm start + projected gradient), and how does it compare to the discrete greedy of Algorithm 6 and to uniform sampling, across the trade-off knob β ?

Setup. We construct a synthetic buffer in the partial-coverage regime, $d = 5$ and $N = 60$ (so N is only a small multiple of d). The target covariance $\mathbf{A}_\pi$ is fixed with a decaying spectrum, and the features $\{\phi_i\}$ (each normalized to the unit sphere, so that *alignment* rather than magnitude drives selection, cf. the intuition trap of §C.2) are drawn in three groups designed to expose the trade-off: (i) an *aligned* group along the top eigendirections of $\mathbf{A}_\pi$ (high leverage); (ii) a *redundant* cluster of near-duplicates along the leading direction (probing the spread-versus-coverage tension); and (iii) a *low-value* group in the small-eigenvalue subspace of $\mathbf{A}_\pi$. The trade-off knob is swept as $\beta = 1/(K(1 - \gamma))$ over several decades of K . We compare: (*baselines*) uniform sampling and the discrete greedy Algorithm 6 with the outer m -scan; (*heuristics*) one-shot leverage sampling $p_i \propto \ell_i$, the Fedorov–Wynn multiplicative-weights iterate, and Fedorov–Wynn + projected gradient on F ; and (*oracle*) a best-of-multistart projected-gradient minimizer of F . Convexity is probed at two levels: random simplex-chord midpoint tests (detects gross non-convexity), and the minimum eigenvalue of the projected tangent Hessian $P \nabla^2 F P$ (with $P = I - \frac{1}{N} \mathbf{1}\mathbf{1}^\top$) sampled over a mixture of interior and *near-boundary* distributions (Dirichlet(α) with small α , and interpolations toward simplex vertices) — boundary points are where the LSTD sandwich is most at risk of non-convexity. Benignness of the landscape is read off from the spread of multistart projected-gradient optima. Solution structure is read off from the effective support $n_{\text{eff}}(p) = 1/\sum_i p_i^2$ and the mass p^* places on each group. [yx:Results, figures, and interpretation to be added in the next revision; numerics in [buffer-selection/numerics/](#).]

Beyond Fixed Target Policies. The development above isolates the buffer-selection mechanism by fixing the target policy π . Re-introducing the actor evolution of Algorithm 3 couples selection with the dual-layer trust-coverage mechanism in two ways. First, the features ϕ_i and the target-aligned leverage scores (C.5) depend on the current reference policy $\bar{\pi}^{(k)}$ through $\phi^{\bar{\pi}^{(k)}}$ and must be recomputed whenever the reference is updated. Second, the gradient-generalization trust metric $\Delta_{k,t}$ in (3.4) is itself defined relative to $\mathbf{A}_k$, which the buffer selection can now actively shape. Both observations point to a natural *adaptive selection* extension in which $S^{(k)}$ is recomputed at each epoch boundary; a full integration with Algorithm 3 is left to future work.